

# Program overview

12-Sep-2023 13:10

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|--------------|------------------------------------------------|
| Year         | 2023/2024                                      |
| Organization | Mechanical, Maritime and Materials Engineering |
| Education    | Master Marine Technology                       |

| Code                                                                              | Omschrijving                                                         | ECTS | p1 | p2 | p3 | p4 | p5 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|------|----|----|----|----|----|
| <b>Master MT 2023</b>                                                             |                                                                      |      |    |    |    |    |    |
| <b>First Year</b>                                                                 |                                                                      |      |    |    |    |    |    |
| <b>Obligatory Courses (25 ECTS)</b>                                               |                                                                      |      |    |    |    |    |    |
| MT44021                                                                           | Motions and Loading of Structures in Waves                           | 5    |    |    |    |    |    |
| MT44030                                                                           | Torsion and Shear in Marine Structures                               | 5    |    |    |    |    |    |
| MT44035                                                                           | Design of Complex Specials                                           | 5    |    |    |    |    |    |
| MT44040                                                                           | Maritime Finance, Business and Law                                   | 5    |    |    |    |    |    |
| MT44050                                                                           | Fundamentals of Marine Engineering                                   | 5    |    |    |    |    |    |
| <b>MT Obligatory Electives (minimum 26 ECTS)</b>                                  |                                                                      |      |    |    |    |    |    |
| <b>Star Electives MT: Select min. 5 courses (min. 23 ECTS)</b>                    |                                                                      |      |    |    |    |    |    |
| ET4117                                                                            | Electrical Machines and Drives                                       | 4    |    |    |    |    |    |
| ME44206                                                                           | Quantitative Methods for Logistics                                   | 5    |    |    |    |    |    |
| ME44305                                                                           | System Analysis and Simulation                                       | 5    |    |    |    |    |    |
| ME44311                                                                           | Advanced Operations and Production Management                        | 5    |    |    |    |    |    |
| ME45042                                                                           | Advanced Fluid Dynamics                                              | 5    |    |    |    |    |    |
| MOT112A                                                                           | Economic Foundation                                                  | 5    |    |    |    |    |    |
| MS43805                                                                           | Material and Fabrication Factors in Marine Structures                | 5    |    |    |    |    |    |
| MT44001                                                                           | Mechatronics in MT                                                   | 5    |    |    |    |    |    |
| MT44006                                                                           | Future Marine Propulsion Systems                                     | 5    |    |    |    |    |    |
| MT44010                                                                           | Non-metallic materials in Marine Structures                          | 5    |    |    |    |    |    |
| MT44025                                                                           | Numerical Ship Hydrodynamics                                         | 5    |    |    |    |    |    |
| MT44061                                                                           | Advanced Course in Resistance and Propulsion                         | 5    |    |    |    |    |    |
| MT44070                                                                           | Shipping Management                                                  | 5    |    |    |    |    |    |
| MT44085                                                                           | Ultimate Strength of Marine Structures                               | 5    |    |    |    |    |    |
| MT44090                                                                           | Fluid - Structure Interaction in Marine Structures                   | 5    |    |    |    |    |    |
| MT44095                                                                           | Design of Advanced Marine Vehicles                                   | 5    |    |    |    |    |    |
| MT44100                                                                           | Internal Combustion Engines A                                        | 5    |    |    |    |    |    |
| MT44105                                                                           | Internal Combustion Engines B                                        | 5    |    |    |    |    |    |
| OE44085                                                                           | Fatigue Strength of Marine Structures                                | 5    |    |    |    |    |    |
| OE44150                                                                           | Marine Hydromechanics                                                | 5    |    |    |    |    |    |
| <b>Star Electives Ethics: Select 1 course (3 ECTS)</b>                            |                                                                      |      |    |    |    |    |    |
| TPM002A                                                                           | Water Ethics                                                         | 3    |    |    |    |    |    |
| TPM404B                                                                           | Technology Entrepreneurship and Internationalization                 | 5    |    |    |    |    |    |
| WM0320TU                                                                          | Ethics and Engineering                                               | 3    |    |    |    |    |    |
| WM0349WB                                                                          | Philosophy of Engineering Science and Design                         | 3    |    |    |    |    |    |
| WM0353TU                                                                          | Climate Ethics                                                       | 3    |    |    |    |    |    |
| WM0375TU                                                                          | Ethics of Technological Risks                                        | 3    |    |    |    |    |    |
| WM0801TU                                                                          | Introduction to Safety Science                                       | 3    |    |    |    |    |    |
| WM1301TU                                                                          | Ethics of Transportation                                             | 3    |    |    |    |    |    |
| <b>Recommended Elective Courses</b>                                               |                                                                      |      |    |    |    |    |    |
| AE4136-22                                                                         | CFD 2: Discretization techniques                                     | 3    |    |    |    |    |    |
| AE4263                                                                            | Modeling, Simulation and Application of Propulsion and Power Systems | 5    |    |    |    |    |    |
| CS4070                                                                            | Multivariate Data Analysis                                           | 5    |    |    |    |    |    |
| ME44125                                                                           | Reliability and Maintenance of Transport Equipment                   | 3    |    |    |    |    |    |
| ME44210                                                                           | Drive & Energy Systems                                               | 3    |    |    |    |    |    |
| ME44300                                                                           | Multi-Machine Coordination for Logistics                             | 3    |    |    |    |    |    |
| ME45001                                                                           | Advanced Heat Transfer                                               | 4    |    |    |    |    |    |
| ME45155                                                                           | Computational Fluid Dynamics for Engineers                           | 5    |    |    |    |    |    |
| ME45160                                                                           | Advanced Applied Thermodynamics                                      | 5    |    |    |    |    |    |
| ME45170                                                                           | Turbomachinery                                                       | 4    |    |    |    |    |    |
| ME46055                                                                           | Engineering Dynamics                                                 | 4    |    |    |    |    |    |
| ME46060                                                                           | Engineering Optimisation: Concepts and Applications                  | 3    |    |    |    |    |    |
| MOT122A                                                                           | Digital Business Process Management                                  | 5    |    |    |    |    |    |
| SC42001                                                                           | Control System Design                                                | 5    |    |    |    |    |    |
| SEN1221                                                                           | Statistical Analysis of Choice Behaviour                             | 5    |    |    |    |    |    |
| <b>Second Year</b>                                                                |                                                                      |      |    |    |    |    |    |
| <b>MT Graduation Project Obligatory (45 ECTS)</b>                                 |                                                                      |      |    |    |    |    |    |
| MT54010-20                                                                        | MT Literature Research                                               | 10   |    |    |    |    |    |
| MT54035                                                                           | MT MSc Thesis                                                        | 35   |    |    |    |    |    |
| <b>MT Project / Electives (choose 1 out of 2, or 15 ECTS of elective courses)</b> |                                                                      |      |    |    |    |    |    |
| MT54000                                                                           | Individual Research Assignment for MT                                | 15   |    |    |    |    |    |
| TUD4040                                                                           | Joint Interdisciplinary Project                                      | 15   |    |    |    |    |    |

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| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

## Master MT 2023

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| <b>Program Title</b>        | MSc Marine Technology (abbreviated to MT-MSc or M-MT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Coordinator</b>  | Peter de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>EC Program</b>           | <p>The Master programme Marine Technology covers two years of study, each with a study load of 60 ECTS. The total programme involves 120 ECTS and comprises cursory modules, assignments and a Master thesis project.</p> <p>Courses (75 ECTS):<br/>These are offered as lectures and/or projects/practicals given simultaneously to all students taking the course. Most courses are assessed by means of a written examination or a (group) report (sometimes including a final presentation). For some courses however assessment takes place by means of an oral examination. There is also a possibility to do an individual research assignment of max. 15 ECTS at the university, at a research institute or in industry (pending approval of an academic staff member of the department as the individual research assignment is not intended as an internship).</p> <p>The set-up of the master with respect to courses is such that there is an obligatory Maritime Technology part (5 courses), an ethics course, a research discipline specific part (min. 5 courses) and an elective part. The research discipline specific parts offers the fundamentals for a further specialisation within Maritime Technology. Through the elective part a student can determine his or her own individual study programme &amp; specialisation to a large extend. Divided over the first 5 quarters (4 quarters in the first year, 1 quarter in the second year), the set-up is as follows:</p> <ul style="list-style-type: none"><li>- 25 ECTS obligatory courses in Maritime Technology (in quarter 1 and 2)</li><li>- 3 ECTS ethics course (choose from list of available ethics courses)</li><li>- 25 ECTS research discipline related courses (a.k.a. star-electives)</li><li>- 22 ECTS "free" electives (courses should be technical and at MSc level)</li></ul> <p>Master Thesis Project (10+35 ECTS):<br/>Each individual student prepares a literature study report and a master thesis of his/her graduation project. The thesis work is further evaluated through an oral presentation by the candidate and an oral examination before a master graduation committee. This committee is composed of at least three scientific staff members, including the thesis supervisor. The graduation committee may also include external examiners from research institutes or from industrial partners. The graduation project is administratively split over 2 courses; the first course is focused on the problem analysis and literature research (resulting in the literature study report), the second course is focused on the solution development, chosen methodology, experiments/simulations, results and conclusions. These parts are graded separately.</p> |
| <b>Prerequisites</b>        | <p>Students from the following studies are directly accepted into the Master:</p> <ul style="list-style-type: none"><li>- BSc Marine Technology (Maritieme Techniek)</li><li>- BSc Mechanical Engineering (Werktuigbouwkunde)</li><li>- BSc Aerospace Engineering (Lucht- en Ruimtevaarttechniek)</li><li>- NLDA BA-MS&amp;T (Profiel MPS-Lvt, MPS-Wtb, C2S-Nav)</li></ul> <p>Students from the following studies need to follow a bridging programme of 30-48 ECTS (see <a href="http://studyguide.tudelft.nl">studyguide.tudelft.nl</a> - 3ME bridging programmes):</p> <ul style="list-style-type: none"><li>- BEng Maritime Technology (HBO scheepsbouwkunde)</li><li>- Maritime Officer Education (Hogere Zeevaart School)</li><li>- BSc Civil Engineering</li></ul> <p>The bridging programme is only available in Dutch and therefore not accessible for foreign students.</p> <p>All other technical BEng and BSc educations will have to follow a bridging program of maximum 60 EC, but which might contain another 20-30 EC in self-study. (see <a href="http://studyguide.tudelft.nl">studyguide.tudelft.nl</a> or <a href="http://doorstroommatrix.nl">doorstroommatrix.nl</a>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Contact for Students</b> | <p>Primary contact: Director of Studies Peter de Vos (to be reached at <a href="mailto:MScMT@tudelft.nl">MScMT@tudelft.nl</a>).</p> <p>Secondary: all other MT staff members.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Introduction 1</b>       | <p>The Master Marine Technology is concerned with all aspects of designing, building and operating high-tech ships (primarily sea-going). It challenges students to integrate various fields of mathematical and physical sciences and to apply fundamental engineering knowledge in a maritime context. Reasons for undertaking such efforts can be e.g. to better assess or predict the behaviour of ships at sea or to optimise the operations of (a) vessel(s). The programme prepares well for a career at a university, in the internationally well-known Dutch shipbuilding and ship-owning industry or foreign maritime industry and/or at world famous research institutes like MARIN and TNO.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Program Goals</b>        | <p>The objective of the Master programme Marine Technology is to educate highly skilled engineering professionals in Marine Technology. With a Marine Technology Master of Science degree one is well prepared for a career as scientist or engineer at an advanced professional level. This level corresponds to the scientific and technological state-of-the-art of a specific discipline within Maritime Technology.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Exit Qualifications</b>  | <p>The Final Qualifications of the MSc Marine Technology can be found in the TEACHING AND EXAMINATION REGULATIONS on the faculty website (<a href="https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/regulations/">https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/regulations/</a>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Structure 1</b>  | <p>The programme structure is already, to some extent, covered in the field "ECTS Program".</p> <p>Additional information w.r.t. tracks and specialisations:<br/>As of academic year 2019/2020 the programme is no longer sub-divided in two different tracks. Instead five research disciplines (or specialisations) in Marine Technology are distinguished (which were there before 2019/2020 as well); each supported by their own research groups.</p> <ol style="list-style-type: none"><li>1. Ship and Offshore Structures (SOS)</li><li>2. Ship Hydromechanics (SH)</li><li>3. Marine Engineering (ME)</li><li>4. Ship Design (SD)</li><li>5. Maritime Operations and Management (MOM)</li></ol> <p>The latter is a merger of the former specialisations Ship Production and Shipping Management.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Program Structure 2</b>  | <p>Marine Technology offers an MSc programme of two years. Each course year is divided into two semesters and each semester consists of two periods (quarters). In this study guide these periods will be referred to as p1, p2, p3 and p4. A period includes seven to eight weeks of lectures, followed by two or three examination weeks. For those courses for which written examinations are held, the student will get at least one opportunity per year to do a resit. Resits are generally held in the first period after the regular period a course is offered. Resits for the examinations held in period 4 are scheduled for half-August (p5).</p> <p>The study load of a course is expressed in European Credits. This is a result of the European Credit Transfer System (ECTS), which encourages acknowledgement of study results between higher education institutions within the European Union. The study load for one educational year is 60 EC. The number of ECTS for a course gives an indication of the weight of (a certain part) of the course. One EC involves approximately 28 hours of study (depending on existing prior knowledge and motivation). These 28 hours include all time spent on the course: lectures, self-study, practicals, examinations, etc.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| <b>First Year</b> |
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|---------------------|-------------------------------------------------------|
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|                                     |
|-------------------------------------|
| <b>Obligatory Courses (25 ECTS)</b> |
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| MT44021                                | Motions and Loading of Structures in Waves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. P. Naaijen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Dr.ir. P.R. Wellens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Summary</b>                         | <p>Motions and loading of structures is about the following topics.</p> <p>Part 1:<br/>Part 1 covers theory of linear excitation and linear motion response of floating structures in waves. Students learn how the linear motion transfer function (RAO) relating ship motion response to incoming wave is computed, including the numerical approach involved and how workability in a given wave spectrum can be assessed including extreme value statistics.</p> <p>In Part 2:<br/>Higher order potential theory and inclusion of non-linear effects in ship motions. Applications to motion of moored ships and to the determination of workability.<br/>The use of Computational Fluid Dynamics (CFD) for the determination of extreme loads on moored FPSO in extreme seas.<br/>One exercise accompanies this part.</p> |   |
| <b>Course Contents</b>                 | <p>Floating structures: wave forces &amp; motions, nonlinear problems, applications</p> <p>Participants completing this segment successfully will have a superior knowledge of and be able to predict the motion of floating bodies in the sea. They will be familiar with first order ship motions in irregular waves as well as drift forces, resulting from nonlinear phenomena. They can also apply this to applications such as station keeping and the determination of offshore workability.</p> <p>Part 1, instructed by Peter Naaijen, is about linear forces, motions and about potential coefficients.</p> <p>Part 2, instructed by Peter Wellens, is about nonlinear forces and the resulting motions.</p>                                                                                                         |   |
| <b>Study Goals</b>                     | <p>Participants who have successfully completed the course will be able to carry out computations at a superior knowledge level involving:</p> <ul style="list-style-type: none"> <li>- linear forces on and resulting motions of ships</li> <li>- nonlinear forces on and resulting ships motions</li> <li>- workability prediction.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | Lectures, exercises and assistance sessions for exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>  | <p>For Part 1, separate lecture notes by Peter Naaijen are available. As an additional reference the Lecture notes "Offshore Hydromechanics" by Journée, Massie and Huijsmans can be used.</p> <p>Exercises "Offshore Hydromechanics" by Journée and Massie.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Prerequisites</b>                   | All participants are required to have successfully completed a basic university level courses in dynamics (mass-spring-damper systems etc.), fluid mechanics and free surface waves before starting this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>Exams:</p> <p>Both part 1 and part 2 provide a summative assessment (exercise) which is not graded.</p> <p>One graded written exam (summative) for Part 1 and Part 2 together at the end of the course.</p> <p>(In case on site examination will not be possible due to COVID19 related circumstances, online examination will be arranged.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Remarks</b>                         | Aggregation of old courses MTM1410 and OE4630 part 2 and 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| MT44030                                | Torsion and Shear in Marine Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5 |
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| <b>Responsible Instructor</b>          | L. Pahlavan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>The course is about torsion of ship and offshore structures.</p> <p>The loading of marine structures consist of {normal, shear} force and {bending, torsion} moment contributions. Torsion becomes particularly important in case of staggered cargo loading conditions and operation in quartering seas, introducing a shear {strain, stress}- and even normal {strain, stress} response in the beam (like) structural members. To be able to establish and calculate the marine structure torsion loading induced response, course subjects include:</p> <p>Free warping torsion: beam {stiffness, strength} characteristics for:</p> <ul style="list-style-type: none"> <li>- solid cross-sections (Saint-Venant theory)</li> <li>- hollow cross-sections: thin walled open (membrane analogy)</li> <li>- hollow cross-sections: thick walled closed</li> <li>- hollow cross-sections: thin walled closed; single-cell (Bredt approximation)</li> <li>- hollow cross-sections: thin walled closed; multi-cell</li> <li>- hollow cross-sections: thin walled mixed (hybrid).</li> </ul> <p>Free warping torsion: beam displacements for</p> <ul style="list-style-type: none"> <li>- solid cross-sections</li> <li>- hollow cross-sections.</li> </ul> <p>Constrained warping torsion: beam {shear, normal} stress for</p> <ul style="list-style-type: none"> <li>- hollow cross-sections: thin walled open</li> <li>- hollow cross-sections: thin walled closed.</li> </ul> <p>Generalised (beam) torsion equation:</p> <ul style="list-style-type: none"> <li>- hollow cross-sections: thin walled open</li> <li>- hollow cross-sections: thin walled closed.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>Theory (fundamentals):</p> <ul style="list-style-type: none"> <li>- to model free warping torsion of beam (like) structural members mathematically and {establish, calculate} the {stiffness, strength} characteristics as well as displacements for {solid, hollow} cross-sections</li> <li>- to model constrained warping torsion of beam (like) structural members mathematically and {establish, calculate} the introduced {shear, normal} stresses for hollow cross-sections</li> <li>- to establish the generalised (beam) torsion equation for hollow cross-sections</li> </ul> <p>Application (engineering):</p> <ul style="list-style-type: none"> <li>- to identify the torsion load carrying members (efficiency) in marine structures at {local, global} level</li> <li>- to {establish, calculate} the torsion loading induced response of marine structures for analysis and design purposes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                | Lectures (using peer education) and assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Assignment reports (80%) and oral exam (20%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| MT44035                                | Design of Complex Specials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. A.A. Kana                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>This course will focus on modern design and engineering approaches and tools for complex marine vessels. This course moves beyond the traditional point based design and the design spiral method and covers both theory and application of modern techniques designed to explore the relationships between requirements elucidation, design generation, design analysis, and design selection. In addition to lecture material, students will also participate in a substantial collaborative design project. Students will develop the preliminary design of a small fleet of offshore wind service vessels for the North Sea in a design team of 6 students taking into account some of the advanced design techniques introduced in the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>All resources, assignments, and deliverables are organized in the course Brightspace page.</p> <p>At the end of the course, the student should be able to:</p> <ol style="list-style-type: none"> <li>1. Identify and explain key features of complex marine vessels, including: vessel missions, operations, tasks, and specific design attributes and principles.</li> <li>2. Describe and compare primary concepts and applications of modern design approaches and tools. Design approaches may include: spiral based design, systems engineering, set based design, concurrent design, modular design, or current TU delft concept ship design research. Design tools discussed may include: Modular Function Deployment, Analytic Hierarchy Process, single and multi-objective optimization, Markov processes, or network theory</li> <li>3. Perform preliminary calculations of modern maritime design tools and techniques.</li> <li>4. Be able to select, apply, and defend the use of one of the modern design approaches and tools to specific design projects or problems.</li> <li>5. Integrate traditional ship design techniques with modern design approaches and tools in the preliminary design of a complex vessel.</li> <li>6. Be able to communicate key aspects and rationale behind complex vessel design both in writing and in a presentation.</li> </ol> |   |
| <b>Education Method</b>                | This course will be given in-person on the TU Delft campus, including a final in-person presentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | The assessment will consist of one or two individual assignments, one midterm report, one final report and presentation, and one peer review evaluation. All assignments and reports will be delivered via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact</b>                         | Contact Austin Kana, a.a.kana@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |



| MT44040                                | Maritime Finance, Business and Law                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. J.F.J. Pruijn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/12/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Summary</b>                         | <p>While technology has the potential to be a game changer, an Engineer is unable to assess its potential if he does not understand the context or environment in which it has to operate. This course will focus on the Legal, Financial and Business (Economic) side of the maritime industry. These three elements form the success criteria for every new innovation; when the law does not allow the use of the innovation when you cannot obtain financing for your new innovation, or when no money can be made with the innovation, it will not be a success. In this course, all three elements will be treated both through guest lectures, assignments and an interactive game designed to test your knowledge and skills in these subjects.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>The course will consist of five elements:</p> <ul style="list-style-type: none"> <li>- A lecture series</li> <li>- Writing a business plan</li> <li>- Writing a year report</li> <li>- The Maritime Business Game</li> <li>- The Grand Annual Meeting</li> <li>- Law Assignment</li> </ul> <p><b>Lecture Series</b><br/>The lecture series filled mostly with guest lectures from industry. Attendance will be taken and counted towards your grade. About nine afternoons will be filled this way. Subjects most likely contain (in random order):</p> <ul style="list-style-type: none"> <li>- Ship Finance</li> <li>- Judging Financial Statements</li> <li>- Ship Insurance</li> <li>- Contract Forms</li> <li>- Maritime Law</li> <li>- Law Enforcement</li> <li>- (IMO) Politics</li> <li>- Sustainability Legislation</li> </ul> <p><b>Writing a business plan</b><br/>Before venturing into the game with your own company an investment proposal, market analysis and strategy (together forming a business plan) will need to be created. Information from both the lectures and the game can be used to show that your plan will be successful. If you manage to convince Mr. Jones of the potential of your plan you will be granted the starting capital to execute it. Both the feasibility and quality of the business plan will be judged.</p> <p><b>The Maritime Business Game</b><br/>The majority of the course time will be spent on testing and honing your skills in strategic, financial and operational management. In the game, you will try to execute your strategy and turn your business (plan) into a success, while dealing with the dynamics the Maritime Industry is known for. Both your success in the game as well as your understanding will count towards your final grade.</p> <p><b>The Grand Annual Meeting &amp; Writing a year report</b><br/>After the game, its time to face your shareholders. You'll provide them with a year report and your plans for the future, both in writing and in a presentation. Did you manage to create the value you promised for Mr Jones and his associates, or will he be greatly disappointed? At the end, the shareholders will speak. Both your presentation as well as your success with the shareholders will count towards your final grade.</p> <p><b>Law Assignment</b><br/>An essay on a predetermined case. Who is to blame according to the law? Write your argumentation using law texts and court decisions for the right interpretation of them.</p> |   |
| <b>Study Goals</b>                     | <p>The student must be able to:</p> <ol style="list-style-type: none"> <li>1. Analyse and interpret financial statements of maritime companies by using at least the following indicators: solvability, liquidity, rentability, working capital, quick ratio, current ratio and debt ratio.</li> <li>2. evaluate an investment by constructing cash flows for the investment and applying Pay-back, Net Present Value, Discounted Cash Flow and Internal Rate of Return techniques to that.</li> <li>3. understands the effects of market movements e.g. exchange rates, freight rates and second-hand value on the position of the company and is familiar with the mitigation options for such dynamics.</li> <li>4. formulate and execute a sound strategy for a maritime company in a dynamic environment</li> <li>5. describe how maritime law is structured, the main legal issues in maritime law, what it entails and how it may affect the competitiveness of maritime companies</li> <li>6. The student is familiar with various common forms of financing for assets, the different costs involved and how the leverage obtained can impact the results of a maritime company.</li> <li>7. Calculate and Discuss the different environmental impact factors for vessels.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                | Lectures, assignments and a Game                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Assessment will be based on seven elements:</p> <ul style="list-style-type: none"> <li>- Lecture attendance</li> <li>- Quality of the Business Plan</li> <li>- Success in the Game (Return on Equity)</li> <li>- Year Results Report</li> <li>- Year Results Presentation</li> <li>- Shareholder Happiness</li> <li>- Law Assignment</li> </ul> <p>Weighing factors will be detailed in Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| MT44050                                | Fundamentals of Marine Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible Instructor</b>          | Dr.ir. L. van Biert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 6/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | <ul style="list-style-type: none"> <li>- Basic mathematics and physics as taught in fundamental (engineering) theory courses in the BSc curriculum of Mechanical Engineering, Aerospace Engineering and Maritime Technology.</li> <li>- Energy conservation law and other thermodynamic principles, basic (fluid) mechanics, (third-order) polynomial functions, ship propulsion systems theory.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>Marine engineering is the field of engineering and science that deals with the design and operation of main and auxiliary systems on board ships and other marine or offshore structures. These systems include mechanical, electrical and other distribution systems. In the course Fundamentals of Marine Engineering the following systems and their main components will be covered:</p> <ul style="list-style-type: none"> <li>- Propulsion systems</li> <li>- Electric power generation and distribution</li> <li>- Auxiliary systems, including cooling water, fuel, lubrication oil and other distribution systems</li> <li>- Safety systems, including fire-fighting and bile-ballast water systems</li> <li>- Hotel systems, including heating, ventilation and air conditioning (HVAC) and refrigeration systems</li> <li>- Waste heat recovery (WHR) systems</li> </ul> <p>The course allows you to develop of a thorough understanding of the working principles of these systems and their main components, which is indispensable for a marine engineer tasked with designing and integrating them. The marine engineer needs to make sure that all systems will operate properly in all environmental conditions that can be encountered by the ship while navigating the seas (e.g. extreme temperatures, rough seas, etc.), while taking into account any additional design requirements for all systems which concern their vulnerability, maintainability, energy efficiency and/or emission of pollutant substances.</p> |   |
| <b>Study Goals</b>                     | <p>After completing this course, you should be able to:</p> <ul style="list-style-type: none"> <li>- Provide an overview of main and auxiliary systems on board ships.</li> <li>- Discuss different ship propulsion and auxiliary system concepts as well as their main technical design considerations.</li> <li>- Apply the working principles of relevant components to properly match suppliers and users in complex (main and auxiliary) systems.</li> <li>- Evaluate how the main components in ship (energy) distribution systems interact in design and off-design conditions.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>Lectures + assignments.</p> <p>In the lectures course material from the course books will be elaborated upon using lecture sheets and interactive Q&amp;A sessions.</p> <p>The optional assignments enables students to obtain experience with the required calculations already during the course period. In this way the assignments support students in developing the required insight in the course contents to effectively solve questions. The assignments will be explained and discussed during dedicated feedback sessions (after students have handed in their solutions).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>  | <p>Prof. J. Klein Woud and Prof. D. Stapersma: Marine Engineering: Design of Propulsion and Electric Power Generation Systems, IMarEST, London.</p> <p>Prof. J. Klein Woud and Prof. D. Stapersma: Marine Engineering: Design of Auxiliary Systems, Shafting and Flexible Mounting, IMarEST, London.</p> <p>Lecture sheets and other course material (made available digitally via Brightspace).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | Exam (open book) + report on obligatory ship power plant assignment (made in groups)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Special Information</b>             | <p>Due to the diversity in backgrounds of students following this course some overlap with BSc courses like "Weerstand, Voortstuwing en Aandrijving 2" in the MT-BSc curriculum and "Stroming en Warmte" in the ME-BSc (WB) curriculum can not be avoided.</p> <p>However, more advanced concepts and/or more applications will be discussed for these overlapping subjects.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                         | Old course code: MTM1416                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

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|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

## MT Obligatory Electives (minimum 26 ECTS)

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| <b>Program Title</b>                | MSc Track Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Coordinator</b>          | Dr. ir. Peter Wellens,<br>T: +31 15 278 88251<br>E: p.r.wellens@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Prerequisites</b>                | <p>The curriculum is designed for students, who have obtained a TU Delft BSc degree Marine Technology. Students who gained their BSc degree in Marine Technology from TU Delft or from one of the WEGEMT universities (Newcastle, Southampton, Hamburg, Berlin and Trondheim) may enter the programme without any restraints. Due to intensive contacts with these universities, their Bachelors programmes are known and have been established to comply with the required entrance knowledge and skills for the Master MT.</p> <p>Students with a Dutch academic BSc degree in Aerospace Engineering, Applied Physics, Civil Engineering, Industrial Design Engineering and Mechanical Engineering also have access to the MT programme without prior selection, but they will have to follow a number of additional courses from the Marine Technology BSc degree. Their total study load will amount increase to a maximum of 150 EC (120 + maximum 30 EC).</p> <p>Students holding academic BSc degrees from a technical university other than those mentioned above, may be accepted to the study programme after an evaluation of the curriculum of their Bachelors programme and their study results. The intake coordinator of the Board of Examiners is responsible for this selection and for the determination of the required additional programme. The selection procedure can result in: admission to the programme, admission to the programme with or without additional requirements of maximum 45 EC (see for procedure above) or no admission. For students coming from universities abroad additional university wide requirements and procedures apply.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Introduction 1</b>               | <p>The development of new concepts for ships and other floating structures and the application of new construction materials require in-depth knowledge of and skills in applying the fundamentals of hydrodynamics and structural strength. This is necessary since the design of novel structures generally cannot be based on past experience and existing concepts alone. Design based on the application of knowledge and skills with respect to first principles is key in this respect. The Science track is aimed at further development of such knowledge and skills, assuming that the basics have been mastered in the Bachelor's phase of the programme.</p> <p>The bulk of the Science track is taken up with the study of mathematics, advanced hydrodynamics, properties of new materials, advanced methods of structural analysis and advanced ship concepts.</p> <p>Within this track the student may choose between the following specialisations.</p> <ul style="list-style-type: none"> <li>- Ship Hydromechanics (SH).</li> <li>- Ship and Offshore Structures (SOS).</li> </ul> <p>If desired, the final-year Master's project may cover both areas.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Exit Qualifications</b>          | <p>The goal of the Masters programme Marine Technology is to educate graduates in Marine Technology to an academic engineering level. The level corresponds to the technological borders of a specific discipline. The graduates are capable:</p> <ul style="list-style-type: none"> <li>- To identify, define and analyse problems, for the solution to which marine-technology principles and techniques can contribute</li> <li>- To develop and to produce a sound solution to the problem</li> <li>- To present these solutions effectively</li> </ul> <p>The graduated Master of Marine Technology in the track Science meets, to a sufficient level, the following qualifications:</p> <ol style="list-style-type: none"> <li>1. Broad and profound knowledge of engineering sciences (applied physics and mathematics) and the capability to apply this knowledge at an advanced level in the disciplines of the track Science.</li> <li>2. Broad and profound scientific and technical knowledge of the disciplines in the track Science and the skills to use this knowledge effectively. The disciplines are Mastered at different levels of abstraction, including a reflective understanding of its structures and relations to other fields, and reaching in part the forefront of scientific or industrial research and development. The knowledge is the basis for innovative contributions to the discipline in the form of new designs or development of new knowledge.</li> <li>3. Thorough knowledge of paradigms, methods and tools as well as the skills to actively apply this knowledge for analysing, modelling, simulating, designing and performing research with respect to innovative systems in Science. with an appreciation of different application areas.</li> <li>4. Capability to independently solve technological problems in a systematic way involving problem analysis, formulating sub-problems and providing innovative technical solutions, also in new and unfamiliar situations. This includes a professional attitude towards identifying and acquiring lacking expertise, monitoring and critically evaluating existing knowledge, planning and executing research, adapting to changing circumstances, and integrating new knowledge with an appreciation of its ambiguity, incompleteness and limitations.</li> <li>5. Capability to work both independently and in multidisciplinary teams, interacting effectively with specialists and taking initiatives where necessary.</li> <li>6. Capability to effectively communicate (including presenting and reporting) about ones work such as solutions to problems, conclusions, knowledge and considerations, to both professionals and non-specialised public in the English language.</li> <li>7. Capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility with regard to sustainability, economy and social welfare.</li> <li>8. Attitude to independently maintain professional competence through life-long learning.</li> </ol> |
| <b>Program Structure 1</b>          | <p>In the Science Track, 25 to 26 EC during the first year are devoted to compulsory subjects which include selected topics from both the hydrodynamic and structural disciplines, as shown in the table under "Obligatory Courses". Another 34 to 35 EC is devoted to specific subjects selected in consultation with the head of the hydrodynamics or structures specializations. The "Recommended Courses" may provide some inspiration in selecting these electives.</p> <p>The second master year consists of an internship and a final master thesis project. The master thesis project may be carried out in-house at the university or in co-operation with industry or research institutes, depending on the student's preference and subject to the availability of positions in industry. Refer to Master Year 2 for more information.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Specialisation</b>               | <p>Ship Hydromechanics<br/>Chair: prof.dr.ir. Riaan van 't Veer</p> <p>Ship and Offshore Structures<br/>Chair: prof.dr.ir. M.L. Kaminski</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Transferring from HBO to TUD</b> | <p>Students with a Bachelors degree from a Dutch polytechnical college (so-called HBO) in ship building, or a degree from a nautical college may be admitted (national BEng degrees), if the student has completed the Bachelors programme within 4 years with good results. These students have to follow a pre-Masters programme that consists of courses from the Marine Technology Bachelors programme, as to enable them to study the Masters programme successfully. These students are admitted to the programme only after having finalised the pre-Masters programme. Together with a few dispensations, the total programme becomes approximately 140 EC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

**Star Electives MT: Select min. 5 courses (min. 23 ECTS)**

| ET4117                                 | Electrical Machines and Drives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible Instructor</b>          | H. Vahedi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Required for</b>                    | AC Machines (ET4121)<br>Control of Electrical Drives (ET4291)<br>Electric vehicles & charging technology (SET3100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior knowledge</b>        | Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | By the end of the course, students must be able to:<br>- Apply Maxwell's equations to magnetic circuits including permanent magnets<br>- Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor<br>- Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines<br>- Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines<br>- Verify the operation of DC, induction and synchronous machine using tests on an experimental setup<br>- Choose the appropriate power electronic converter and method for speed control and starting of the machine |   |
| <b>Education Method</b>                | - Classroom lectures and problem solving workshops,<br>- Demonstrations using machines and animation in the classroom,<br>- Online exercises with self-evaluation,<br>- Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6<br>Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | The final grade of the course consists of the following components:<br>- Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets.<br>- Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course.<br><br>Repair possibilities:<br>Written exam: Resit in next quarter<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                         | In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups.<br><br>If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| ME44206                                | Quantitative Methods for Logistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. B. Atasoy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Ir. M.B. Duinkerken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>The course contains of two parts; roughly 7 weeks each.</p> <p>In the first part of the course an introduction to basic operations research techniques will be presented:<br/> - linear programming (simplex, duality etc.)<br/> - integer programming (IP, BIP, MILP etc.)<br/> - basic probability and statistics; brief overview of models to deal with uncertainty<br/> This part of the course will be largely be based on the first chapters of the book by Hillier and Lieberman.<br/> During the first quarter students must complete an individual practical assignment where they develop a mathematical model and implement it using Python together with Gurobi as a solver.</p> <p>In the second part of the course several common quantitative methods for transportation and logistics are presented.<br/> - networks and routing: network flow problem, facility location problems, transportation problem, transshipment problem, vehicle routing problem and other standard problems; standard algorithms i.e. the branch and bound method.<br/> - queueing theory: overview of basic models and results.</p> <p>During the second quarter students must complete a practical assignment in groups on a case study with a transportation network again using Python.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of this course the students will be able to:</p> <ul style="list-style-type: none"> <li>- Explain the importance and use of operations research techniques to analyze and improve systems (i.e. queuing, simulation, forecasting, routing, scheduling)</li> <li>- Formulate mathematical models of real-life problems (e.g., resource allocation, production planning, transportation) with operations research techniques</li> <li>- Verify and validate mathematical models</li> <li>- Evaluate operation research techniques based on strengths and weaknesses (in terms of accuracy and efficiency) for improving real-life problems</li> <li>- Apply basic operation research techniques to small scale problems as well as realistic case studies</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>Flipped classroom approach: The students are expected to be prepared by studying the online material that is indicated every week.</p> <p>Lectures in two parts: First 45 min course content, Second 45 min assignment<br/> Course content includes a short intro and exercises related to the subject.<br/> Assignment part will be dedicated to introduction of the software (Python+Gurobi), introduction of the assignments and any further needed assistance and feedback.<br/> It is strongly recommended to bring a laptop to the lectures.<br/> There will be also additional reading material that will be provided during the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Books</b>                           | Hillier & Lieberman. Introduction to Operations Research. 10th Edition, McGraw-Hill, ISBN 9781259253188                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>Practical Q1 - 25% (individual: Mathematical Model, Python implementation and report)<br/> Practical Q2 - 25% (group: Mathematical Model, Python implementation and report)<br/> Exam Q2 - 50%.</p> <p>For all 3 parts a minimal grade of 5 is required.<br/> The exam will be given at the end of Q2, the resit for the exam in Q3<br/> Both assignments are only given once a year; with a limited option to resubmit if the grade is insufficient. The highest grade for a resubmission is 6.</p> <p>In case of measures resulting from COVID-19, the prescribed assessment will be an online examination. If the opportunity arises for examinations on campus, the method of assessment will be on campus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Remarks</b>                         | TIL-students: please be aware that there is an overlap in content between CIE4835 (elective T&P 3.1) and ME44206 (compulsory TIL). You are only allowed to choose one of the courses in your degree program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| ME44305                                | System Analysis and Simulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Ir. M.B. Duinkerken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | F. Schulte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Required for</b>                    | Multi-Machine Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Basic programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Parts</b>                           | Systems and Data Analysis<br>Discrete-Event and Agent-Based Simulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | The course is an introduction to "Problem Engineering"; how to formulate a well-defined problem in a complex situation. The course introduces a systems approach to define the elements and structure of a problem situation and the process approach of discrete simulation to understand and quantify the time-dependent relations (behavior)                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | Learning objectives:<br>- Describe all relevant elements and their relations for at least one complex industrial or transportation system (e.g., cyber-physical production systems) using a combination of at least two conceptual models: process models, system models, steady-state models;<br>- Collect data for the described system and analyse the system using the collected data and introduced data analysis methods;<br>- Design industrial and transportation systems and processes applying an established innovation model;<br>- Transform the developed conceptual models into a process-oriented simulation model, using a suitable language (e.g., Delphi/Tomas/Lazarus or Python/Salabim/Simpy). |   |
| <b>Education Method</b>                | lectures + group assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Computer Use</b>                    | Windows laptop with Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | "The Delft Systems Approach, Analysis and Design of Industrial Systems", 2008, H.P.M.Veeke, J.A.Ottjes, G. Lodewijks, Springer, ISBN 978-1-84800-176-3<br>Lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | Report of group assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Hours</b>                      | 80 for assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Tags</b>                            | Abstract<br>Broad<br>Challenging<br>Design<br>Industry<br>Logics<br>Modelling<br>Process<br>Programming<br>Programming concepts<br>Prototyping<br>Stochastics<br>Transport & Logistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Percentage of Design</b>            | 50%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Design Content</b>                  | Logistic systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |



| ME44311                                | Advanced Operations and Production Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | A. Napoleone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Ir. M.B. Duinkerken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>        | To enroll in the course, students must have a BSc in Engineering. The course is especially for students following TIL, MME and SDPO.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>The course requires high engagement and commitment, and includes both self-study and group work. Students are expected to be willing to work in teams, to collaborate and use social and project management skills.</p> <p>The course revolves around different topics of operations and production management, including: system analysis and process mapping, modularity in product design and process design, smart manufacturing and Industry 4.0, reconfigurability and adaptability in operations management.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | <p>With this course, the student is expected to develop a structured approach to address operational problems using the methods provided in the course and/or methods identified with individual study.</p> <p>After successfully completing this course, students will be able to:</p> <ul style="list-style-type: none"> <li>Describe the fundamental concepts of operations and production management</li> <li>Search for and use methods to solve operational problems in different applications</li> <li>Have a structured approach to manage familiar as well as unknown operational problems</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Method</b>                | <p>Problem-based and interest-driven approach.</p> <p>PART ONE in Q3: (i) Plenary lectures / Cases by industrial speakers AND (ii) Individual study and deepening of one/more operational topics of interest. Attending lectures is required to acquire the fundamentals of production and operations management. Active participation during plenary lectures is strongly advised. At the end of Q3, students submit individual reports (one report per student).</p> <p>PART TWO in Q4: Group projects (up to 6 students per group) with case companies AND Group supervision (on demand). In Q4, students work in groups at different projects with case companies. The course leader can assist students in finding a company. Groups work autonomously and ask for supervision when needed. At the end of Q4, students submit group reports that include multiple analyses of operational problem at the case company and the solution to the problem. Q5 is available groups if preferable for time constraints.</p> |   |
| <b>Assessment</b>                      | The assessment is based on the evaluation of the individual report (Q3) and group report (Q4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Judgement</b>                       | <p>The report is judged by Dr A. Napoleone</p> <p>The final grading considers the individual report (50%) and the group report (50%)The grading is based on a Rubric.</p> <p>Results can be discussed in feedback sessions (on demand)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact</b>                         | a.napoleone@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| ME45042                                | Advanced Fluid Dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5 |
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| <b>Responsible Instructor</b>          | Dr. A.J.L.L. Buchner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. D.S.W. Tam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 1<br>2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | Undergraduate level course in thermo-fluids (WB2542)<br>Knowledge of vector analysis, multivariable and differential calculus is essential for this course (WBMT1050, WBMT2048)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | This course surveys the principal concepts and methods of fluid mechanics. Topics include control volume analysis for mass and momentum, the derivation of Navier-Stokes equations from mass and momentum conservation for incompressible flows, boundary conditions at fluid/fluid and fluid/solid interfaces, dimensional analysis, interfacial tension, inviscid flows, circulation and vorticity, lift and drag, boundary layers, lubrication theory, classical instabilities and waves. Classical solutions of the Navier-Stokes equations are derived and Concepts are illustrated through practical examples from engineering, geophysics and biological fluid dynamics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>At the end of this class, the student is able to formulate and to use the fundamental governing equations of incompressible, laminar fluid mechanics in classical flow regimes, and to apply these to a new problem in flow technology.</p> <p>More specifically, the student must know how to:</p> <ol style="list-style-type: none"> <li>1. formulate the conservation equations for mass and momentum</li> <li>2. derive the governing equations of motion for an incompressible flow. (Starting from the conservation equations for mass and momentum, the constitutive equation for a Newtonian fluid, and the appropriate boundary conditions at a fluid/solid and fluid/fluid interface)</li> <li>3. compute the stress tensor and the aero/hydrodynamic forces on a mechanical system</li> <li>4. derive the vorticity equation and the consequence on the generation and transport of vorticity in 2D (Kelvin circulation theorem)</li> <li>5. perform a dimensional or scaling analysis to identify the dominant terms in the governing equations. Also, use the non-dimensional numbers introduced in class to determine the flow regime (Reynolds, Strouhal, Froude, Bond, Capillary, Weber)</li> <li>6. simplify the Navier-Stokes equations for the case of an inviscid flow (the Euler equations), a boundary-layer, a unidirectional flow, a creeping (Stokes) flow and in the lubrication limit.</li> <li>7. apply appropriately the different forms of the Bernoulli equation</li> </ol> <p>The student must be able to derive classical solutions to the Navier-Stokes equations discussed in class. The student must be able to determine whether these classical solutions can be applied to a new problem and use them to compute the flow fields. These solutions include:</p> <ol style="list-style-type: none"> <li>8. For ideal flows: a source, a sink, an irrotational vortex, a dipole, corner flows and the superposition of any combination of these using complex potentials (e.g. the 2D irrotational flow around a cylinder and the interpretation of Lift (Joukowski theorem) and Drag (d'Alembert Paradox))</li> <li>9. 3D axisymmetric irrotational flow around a sphere for steady and unsteady flows (added mass)</li> <li>10. For viscous flows: Unidirectional laminar flows, Couette and Poiseuille flows, flow in a lubrication layer</li> <li>11. Blasius solution for the boundary layer over a flat plate</li> <li>12. Waves and instabilities in fluid mechanics, gravity waves, Kelvin-Helmholtz instability</li> </ol> <p>When facing a new problem in fluid dynamics, the student should be able to:</p> <ol style="list-style-type: none"> <li>13. Write all the relevant boundary conditions</li> <li>14. Use the appropriate formulation of mass and momentum transport (at the control volume level or at the differential level i.e. Navier Stokes equations)</li> <li>15. Simplify the governing equations using a dimensional/scaling analysis</li> <li>16. Solve the governing equations for simple flows, when analytically possible</li> <li>17. Use well-known solutions derived in class to model flows appropriately for the new problem.</li> <li>18. Model the effect of aero/hydrodynamic forces on the dynamics of a simple mechanical system</li> </ol> |   |
| <b>Education Method</b>                | Lectures: 3 hours/week<br>Recitations: 1 hour/week<br>Office Hours: 1 hour/week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Books</b>                           | Fluid Mechanics<br>Sixth Edition<br>Kundu, Cohen and Dowling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | Midterm Written Exam (35%)<br>Final Written Exam (65%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact</b>                         | email: a.j.buchner@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| MOT112A                                | Economic Foundation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5 |
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| <b>Module Manager</b>                  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-responsible for assignments</b>  | Dr. C.W.M. Naastepad                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Required for</b>                    | Bachelor students are excluded from participation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Summary</b>                         | <p>Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works, why it sometimes does not work as well as we would like, how its performance can be improved by regulation and policy, and how over time it changes and develops. The course does not assume that the student has any prior knowledge of economics. The course has two parts: (1) microeconomics, which covers the standard topics of supply and demand, forms of market competition, imperfect competition, efficiency, labour markets and technological progress by firms; and (2) macroeconomics, which examines aggregate demand, growth, welfare, inflation, unemployment and globalization. In both parts, special attention is given to (economic factors explaining) the extraordinary technological dynamism and productivity growth of market economies and to governments role in improving micro- and macroeconomic performance.</p> <p>Microeconomics (2,5 ECTS) focuses on markets, and takes both the firm as the government as principal actors. How are firms' attempts to realize their objectives (in terms of profits, sales or market shares) influenced by demand, technology and market structure? How can public regulation improve market performance which falls below norms considered acceptable? We identify important market failures, due to which markets under-perform, and specifically investigate firm behaviour in (realistic) oligopolistic markets. Macroeconomics (2,5 ECTS) deals with the workings of the economy as a whole, why the market system often underperforms, and how macroeconomic performance can be improved by fiscal and/or monetary policy. Attention will be given to conflicts between achieving different goals of macroeconomic policy at the same time, and to the ways in which globalization affects the capacity of governments to regulate their economies. The answer to this question holds major lessons for public policy.</p> |   |
| <b>Course Contents</b>                 | <p>Microeconomics: demand and supply; consumer behavior; market competition; perfect competition; monopoly; oligopoly; market failures; externalities; public goods; regulation; efficiency; welfare; production; profit maximization; choice of technique; technological progress; game theory; dominant strategy; labour markets; employment; worker-management relations.</p> <p>Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; neoclassical economics; Keynesian economics; unemployment; fiscal policy; monetary policy; money creation; crowding out; stabilization policies; labour market regulation; technology policy; foreign direct investment; financial markets.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | <p>At the end of this course, students will be able to:</p> <ol style="list-style-type: none"> <li>1. analyze firm behavior in markets of perfect competition, monopoly, and oligopoly as well as the impact of firm behavior on economic welfare.</li> <li>2. identify and appraise the rationale, scope and limits of public market regulation (in situations of "imperfect markets" and/or "market failures").</li> <li>3. derive and document the impact(-s) of regulation on the static efficiency and dynamic efficiency of markets.</li> <li>4. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness.</li> <li>5. critically examine and reflect on the existing alternative theories and models.</li> <li>6. understand how money is created and how the financial sector works, and how this affects the real economy.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>                | Lectures. Students are expected to prepare for each class by reading the indicated literature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Books</b>                           | Lecture Notes, Question sets and papers/articles - all made available on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>In normal circumstances, the exam is a closed-book electronic (in-person, on campus) multiple-choice examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.5. Students will be offered two opportunities to do the exam in an academic year. In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book &amp; timed multiple-choice examination of 3 hours.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Special Information</b>             | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| MS43805                                | Material and Fabrication Factors in Marine Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Y. Gonzalez Garcia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ir. M.J.M. Hermans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr.ir. R.H. Petrov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>This course is focused on the application of metals in marines structures. Attention is paid to the structure and properties of steels and light alloys as well as their corrosion resistance. The weldability of high-strength steels is addressed and factors influencing weld joint design and design for welding are considered, together with assessment considerations. The fundamental aspects of corrosion mechanism, corrosion monitoring and control will be introduced. The general outline of the course (number of lectures):</p> <ul style="list-style-type: none"> <li>- Basics of metal structures, phase diagram, phase transformation of metals (2)</li> <li>- Influence of alloying elements on steel structures (1)</li> <li>- Steel production basics (1)</li> <li>- Rolling technology (2)</li> <li>- Al and Al alloys (1)</li> <li>- Welding process &amp; Solidification (2)</li> <li>- Joining dissimilar materials (2)</li> <li>- Additive manufacturing (1)</li> <li>- Corrosion: Thermodynamic and Kinetics (2)</li> <li>- Corrosion types (1)</li> <li>- Corrosion steels and Light-alloys (1)</li> <li>- Corrosion Protection and Control (1)</li> <li>- Cathodic protection (2)</li> </ul> |   |
| <b>Course Contents Continuation</b>    | <p>The group assignment will focus on a case selected by the students. Based on the specific requirements of the chosen case, the group must (i) recommend the most appropriate material(s) and production procedure, (ii) assess the risk for mechanical failures and propose possible mitigation technique(s) and (iii) evaluate the corrosion susceptibility and recommend method(s) for protection. Each group will deliver a written report. Example of cases:</p> <ul style="list-style-type: none"> <li>- A helideck on an offshore platform</li> <li>- The boom of a high lift capacity offshore crane</li> <li>- A new oil pipeline to be laid at 3-km water-depth</li> <li>- An lightweight alloy ship hull</li> <li>- Liquified natural gas storage tank</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <ul style="list-style-type: none"> <li>- Describe the main factors governing the structure and properties of metals</li> <li>- Identify joint designs typically employed in fabrication</li> <li>- Determine design factors influencing welded joint design and joint performance</li> <li>- Describe corrosion principles and the correlation with material properties</li> <li>- Identify the main corrosion types frequently found in marine structures</li> <li>- Determine appropriate corrosion protection methods based on their underlying behaviour and mechanisms</li> <li>- Select criteria for corrosion control programs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Method</b>                | Lectures and recommended reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>The metal fabrication part of the course is addressed through readers. The corrosion part of the course is based on selected book chapters that will be available during the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Examination (60%) and Assignment (40%). The exam is in closed-book written format. Students are required to have a minimum score of 6 in both the examination and the assignment to pass this course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| MT44001                                | Mechatronics in MT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | A. Coraddu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc courses on control engineering (incl. Proportional-Integral-Derivative control), hydromechanics, marine engineering, and signal analysis and processing.<br>Software Skills: Matlab/Simulink.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>Overview:<br/>Unlock the fascinating intersection of mechanics, electronics, and information technology in maritime settings through this cutting-edge course. Our main objective is to guide you in designing a Dynamic Positioning (DP) system a complex mechanism to stabilize a model boat against various natural disturbances like waves, wind, and currents.</p> <p>What You'll Learn:<br/>Integrated Systems Approach: Explore the synergy between mechanics, electronics, and IT in maritime projects.<br/>Dynamic Positioning (DP) System: Delve into designing and constructing a DP system to stabilize a model boat against environmental variables.<br/>Hands-on Experience: Use the Matlab/Simulink environment to create your DP model.<br/>Real-world Implementation: Transition from theoretical design to hands-on testing in our state-of-the-art lab.<br/>Critical Evaluation: Assess the strengths and weaknesses of your DP model in terms of performance and reliability.</p> <p>Core Components:<br/>Marine Engineering: Understand vessel design, propulsion, and other marine systems fundamentals.<br/>Control Engineering: Gain in-depth knowledge of control theory, including PID controls essential for dynamic systems.<br/>Digital Signal Processing: Master techniques for analyzing, interpreting, and manipulating signals critical for effective control systems.</p> <p>Why Take This Course?<br/>Interdisciplinary Learning: Equip yourself with a unique blend of skills from mechanics to digital signal processing, setting a robust foundation for your engineering career.<br/>Hands-on Application: Bridge the gap between theoretical knowledge and real-world applications through lab experiments.<br/>Career Relevance: Prepare for an ever-evolving marine, aerospace, and control systems engineering job market.<br/>Holistic Understanding: Gain a 360-degree perspective on maritime mechatronic systems, with the DP system serving as a quintessential example.<br/>Performance Analysis: Acquire the skills to critically evaluate and optimize complex engineering systems for real-world challenges.</p> <p>By the end of this course, you'll have a working DP model and a comprehensive understanding of how to design, analyze, and evaluate complex maritime mechatronic systems.</p> |   |
| <b>Study Goals</b>                     | <p>1 - Develop an optimal DP solution for a given ship by understanding the theory and practice.<br/>2 - Apply the basics of computer-based modeling, measuring, and control systems.<br/>3 - Apply basics of digital signal processing relevant to mechatronics systems (sampling, filtering, signal resolution, A/D and D/A conversion).<br/>4 - Describe the basic concepts of mechatronics and the working principles of sensors and actuators.<br/>5 - Analyse and interpret the performance of mechatronics systems.<br/>6 - Evaluate the differences between the theory and practice of control systems.<br/>7 - Evaluate the strengths and weaknesses of mechatronic systems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals continuation</b>        | <p>After successfully completing the course, students will be able to:</p> <ul style="list-style-type: none"> <li>- Describe the basic principles of mechatronics</li> <li>- Describe sensors and actuators in general and the ones specific to the project,</li> <li>- Apply basics of signal processing relevant to mechatronic systems (sampling, filtering, signal resolution, A/D and D/A conversion),</li> <li>- Use and tune computer-based measuring and control systems,</li> <li>- Analyse and interpret the performance of a miniature Dynamic Positioning system regarding theoretical and practical aspects,</li> <li>- Evaluate the strengths and weaknesses of a mechatronic system that the students set up.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Lectures + Assignments.</p> <p>In the lectures, course material from the course books will be elaborated upon using lecture sheets and interactive Q&amp;A sessions.</p> <p>The assignments enable students to obtain experience with the required calculations during the course period. In this way, the assignments support students in developing the required insight into the course contents to submit the deliverables.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Books</b>                           | <p>Bolton, W. (2003). Mechatronics: electronic control systems in mechanical and electrical engineering. Pearson Education.<br/>Karnopp, D. C., Margolis, D. L., &amp; Rosenberg, R. C. (2012). System dynamics: modeling, simulation, and control of mechatronic systems. John Wiley &amp; Sons.<br/>Cetinkunt, S. (2015). Mechatronics with experiments. John Wiley &amp; Sons.<br/>Fossen, T. I. (1994). Guidance and control of ocean vehicles. John Wiley &amp; Sons Inc.<br/>Wadoo, S., &amp; Kachroo, P. (2016). Autonomous underwater vehicles: modeling, control design and simulation. CRC Press.<br/>Karris, S. T. (2006). Introduction to Simulink with engineering applications. Orchard Publications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Reader</b>                          | <p>Reader (available on Brightspace).<br/>Week assignments (available on Brightspace).<br/>Additional material supplied on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on their performance with respect to the following deliverables:<br/>D1: Demonstrate a selected DP solution using a working (online) control system computer model - 30%.<br/>D2: Demonstrate multiple DP solutions using a working computer model (offline) - 30%.<br/>D3: Document your solutions in a report, discuss the validity of the offline model as a representation of the online system, and reflect on the learning process - 40%.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Remarks</b>                         | <p>The maximum capacity of MT44001 is 35 students per period. The reason is the availability of the flume tanks, boats, required charging time, etc.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact</b>                         | Dr Andrea Coraddu - a.coraddu@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| MT44006                                | Future Marine Propulsion Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.C. van Terwisga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Ir. J.S. Grevink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | As this course bridges between "Ship Resistance and Propulsion" and "Marine Engineering", a basic understanding of the principles of both fields is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Summary</b>                         | <p>In the course Marine Propulsion Systems the characteristic of ship propulsors are discussed in relation to their driving machines as well as the ships they operate on. The course intends to teach students how to make the right decision when selecting a propulsor and propulsion system with respect to efficiency, operational possibilities, flexibility, economy (CAPEX and OPEX), the impact on the environment and the interaction with local conditions.</p> <p>During the course, the physical principles and theory are translated into practical situations and the challenges designers face to select optimal propulsion for ships and other floating structures. Furthermore, students are introduced to on-going research about and related to ship propulsion systems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>In Marine Propulsion Systems the characteristics of ship propulsors are discussed in relation to their prime movers as well as the ships they operate on. It combines insights gathered from the fields of ship hydromechanics and marine engineering into practical situations which arise during the design and operation of both ships and ship systems. Different aspects of ship propulsion systems are discussed and include typical system configurations, hydromechanics, propulsive efficiency, system efficiency optimization, cavitation, pressure pulses, noise and vibrations, model test analyses, influence of various types of prime movers, selection criteria, static and dynamic behaviour, control strategies, free sailing performance, manoeuvring, dynamic positioning, high speed ship propulsion, mechanical design of propulsors, classification requirements, cost analyses, auxiliary systems and relevant environmental issues.</p> <p>This course discusses the various types of propulsors and their characteristics:</p> <ul style="list-style-type: none"> <li>- Fixed Pitch Propellers</li> <li>- Controllable Pitch Propellers</li> <li>- Marine Gearboxes</li> <li>- Steerable Thrusters</li> <li>- Podded propulsors</li> <li>- Transverse Tunnel Thrusters</li> <li>- Manoeuvring and rudders</li> <li>- Waterjets</li> <li>- Voith Schneider Propellers</li> </ul> <p>As energy saving is in many cases the dominant selection criterion of propulsion systems, extra attention is given to various kinds of energy saving devices and their basic working principles. These energy saving devices typically focus on reclaiming or preventing some of the losses that occur during the propulsion of ships.</p> <p>The application of nozzles benefits propulsion mainly during manoeuvring and slow ship speed operation, but it may contribute to the ships performance also for free sailing applications. The various choices for nozzle types are discussed and compared in this course.</p> <p>Another aspect that is discussed is the proper matching of propeller and the various types of driving machines, mainly 4-stroke and 2-stroke diesel engines as well as electric motors. What are the consequences of the choices made for the propulsion drivetrain with regards to fuel economy, manoeuvrability, comfort, dynamic behaviour as well as initial and operational costs.</p> <p>Rather than a defined set of competences, this course will familiarize students with sometimes complex interaction between the various set of variable boundary conditions and the consequences of the design choices to a vessels propulsion performance. This course will create a mind-set with which you can assess analytically the merits and drawbacks of different choices of propulsors.</p> <p>Furthermore, "hot topics" in ship propulsion system research will be discussed in "research lectures". Topics may include:</p> <ul style="list-style-type: none"> <li>- Wind-assisted propulsion</li> <li>- Air lubrication</li> <li>- Advanced pitch / propulsion control</li> <li>- Alternative materials and / or alternative fuels</li> <li>- Alternative power generation (e.g. fuel cells).</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After this course the student is able to</p> <ul style="list-style-type: none"> <li>- compare various propulsion systems in relation to performance, total propulsion system efficiency, dynamics, maneuverability, comfort, design and costs.</li> <li>- compare the impact of propeller design point choices in different working points of a propeller, using knowledge of prime mover and propeller open water diagrams.</li> <li>- explain the various propulsor influence factors for optimal free sailing, maneuvering or Dynamic Positioning performance.</li> <li>- understand and explain the working principles and characteristics of: <ul style="list-style-type: none"> <li>o Fixed Pitch Propellers</li> <li>o Controllable Pitch Propellers</li> <li>o Marine Gearboxes</li> <li>o Rudders</li> <li>o Steerable Thrusters</li> <li>o Podded propulsors</li> <li>o Transverse tunnel thrusters</li> <li>o Waterjets</li> <li>o Voith Schneider Propellers</li> </ul> </li> <li>- describe shortly topics in ship propulsion system research and how they relate to each other</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Lectures + 3 optional assignments; support and feedback will be organized.</p> <p>Lectures provide not only the slides, but also additional information that is only conveyed orally. It is highly recommended to attend all lectures and take notes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Handouts of the lectures, additional reading material will be provided on brightspace</p> <p>Supporting literature:</p> <p>Prof. J. Klein Woud and D. Stapersma: "Design of propulsion and electrical power generation systems", IMarEST, London.</p> <p>Prof. G. Kuiper: "Resistance and Propulsion"</p> <p>John Carlton: "Marine Propellers and Propulsion"</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Exam + essay / mini-paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

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| <b>Department</b> | 3mE Department Maritime & Transport Technology |
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| <b>MT44010</b>                         | <b>Non-metallic materials in Marine Structures</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>5</b> |
| <b>Responsible Instructor</b>          | L. Pahlavan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Course Contents</b>                 | In this course you will learn about fiber-reinforced composite (FRC) materials and structures. Applications of FRCs are rapidly growing in the maritime sector due to their light-weight and superior corrosion resistance compared to their metallic counterparts. For those who have interest in structural design and analysis, it is important to understand how these materials can be designed, manufactured and recycled, applied to modern maritime of offshore structures, and inspected/monitored. MT44010 includes two parts: (i) fundamentals of FRC materials; and (ii) analysis and assessment of structures made of FRCs. The first part of the course introduces the manufacturing processes, recycling methods, microstructural characteristics, and mechanical behavior of composites highlighting the process-microstructure-property relationship. The second part focuses on the analysis of structures made of FRCs, and fundamentals of the conventional and advanced measurement methods for assessment of composites. The course contains two project assignments, and an experimental demonstration session. |          |
| <b>Study Goals</b>                     | <p>Upon completion of the course, students should be able to:</p> <ol style="list-style-type: none"> <li>1. Describe manufacturing and recycling processes for composite materials</li> <li>2. Relate microstructure and properties of composite materials</li> <li>3. Estimate strength of composite lamina from both micro- and macro-mechanics theories</li> <li>4. Analyze elastic behavior of composite lamina and laminates using Classical Laminate Theory</li> <li>5. Assess bending, vibration, and buckling behavior of structures made of composite material</li> <li>6. Select suitable measurement techniques for assessment and monitoring of FRCs</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Education Method</b>                | Lectures, project assignments, and self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Literature and Study Materials</b>  | <p>Daniel and Ishai (2006), Engineering Mechanics of Composite Materials, Oxford university press<br/> Reddy, J. N. (1997), Mechanics of Laminated Composite Plates. CRC<br/> Kassapoglou, C. (2013), Design and Analysis of Composite Structures, Wiley</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Assessment</b>                      | <p>Project assignments (80%)<br/> Presentation of project results and oral assessment (20%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |

| MT44025                                | Numerical Ship Hydrodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. H.J. de Koning Gans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>          | Dr.ir. I. Akkerman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>Numerical methods for ship hydrodynamics will be discussed and used to predict some relevant ship hydrodynamic phenomena. This will cover both inviscid potential methods for diffraction and wave making prediction and viscous methods for the prediction of damping.</p> <p>The course will cover theoretical development of panel methods and methods for predicting viscous flow. Using mathematical concepts such as Greens function theory and weak formulations and there use in constructing numerical fluid models. More practical aspects such as mesh generation and error estimation will also be covered. Finally, the students will be familiarised with remote computing on a dedicated linux cluster.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of the course the students will be able to:</p> <ol style="list-style-type: none"> <li>1 explain the principles behind panel codes</li> <li>2 explain the principles behind viscous flow codes</li> <li>3 describe different meshes and mesh generation techniques</li> <li>4 make a mesh suitable for ship hydrodynamic simulation</li> <li>5 select appropriate run parameters for ship hydro hydrodynamic simulation</li> <li>6 perform computation on remote cluster</li> <li>7 analyse computed results and draw appropriate ship hydrodynamic conclusions</li> </ol>                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures and self study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Computer Use</b>                    | An account on a in house cluster will be provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Koning Gans, Dr. Ir. H.J. de "Numerical Methods in Ship Hydromechanics"</p> <p>Koning Gans, Dr. Ir. H.J. de "Manual of Numerical Methods in Ship Hydromechanics"</p> <p>Koning Gans, Dr. Ir. H.J. de "From source point to panel methods"</p> <p>Optional books:</p> <p>"Low Speed Aerodynamics From Wing Theory to Panel Methods", Katz, J. and Plotkin, A.</p> <p>"Practical Ship Hydrodynamics", Bertram, V</p>                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | <p>Two written reports both counting for 50% of the total grade. Each report should have agreed of at least 5.8. A resubmit is capped at 70% to factor in the feedback.</p> <p>Report 1:</p> <p>Analyse the diffraction in waves and wave making resistance in flat water using potential methods</p> <p>Report 2:</p> <p>Analyse of a planning or slamming ship using a viscous flow solver. Or some equivalent simplified flow problem with a prominent free-surface effect.</p>                                                                                                                                                                                                                                            |   |
| <b>Remarks</b>                         | Old course code: MTM1411                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |



| MT44061                                | Advanced Course in Resistance and Propulsion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. T.J.C. van Terwisga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | Basic knowledge (BSc level) of the physics, experimental and computational techniques in Fluid Dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>This course aims to link education with state of the art research in the field of resistance and propulsion, through a combination of lectures and scientific papers.</p> <p>Use of CFD in holistic optimization of the integrated system: hull form, propeller, rudder, possible Wind Assisted Propulsion and Energy Saving Device (ESD)</p> <p>Uncertainty assessment in experiments and CFD computations</p> <p>Frictional Drag Reduction by Air Lubrication</p> <p>Principles of Energy Saving and Energy Saving Devices</p> <p>Predicting Cavitation Nuisance; pressure fluctuations, cavitation erosion and noise</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>Basic knowledge or an interest in CFD and fluid dynamics is expected.</p> <p>This course aims to link education with research.</p> <p>The more detailed objectives of this course are to train the student to:</p> <ol style="list-style-type: none"> <li>1. Reproduce the outline of the latest developments in the field of Resistance and Propulsion Hydrodynamics. For topics, see course contents</li> <li>2. Analyse a hydrodynamic problem in the resistance and propulsion area and subdivide it into well defined subproblems that can be analysed with state of the art knowledge and tools</li> <li>3. Select the appropriate theory or tool (either numerical or experimental) for an analysis of the identified problem</li> <li>4. Understand, reproduce and present to an audience, the main lines in a contemporary publication from the field of Propulsion and Resistance hydrodynamics</li> <li>5. Understand, interpret and react to questions from the audience and the lecturer and in doing so, stimulate the scientific debate</li> </ol>                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>In the first quarter of the course, a series of lectures on selected topics (see Course Contents) is given. To ensure an active role of the student in acquiring the knowledge, two homework exercises are offered. These exercises can be made with the material that is presented during lectures.</p> <p>The homework assignments will address results from a Numerical Analysis of the flow about a ship, propeller or part of it. Based on these results, the student is asked to interpret the results, given the modelling that was used. Numerical and/or experimental uncertainties will also be addressed.</p> <p>In the second quarter of the course each student is requested to present a scientific paper in groups of two students. The papers are related to the topics addressed in the first quarter of the course. Approved scientific papers will be made available from which the students can select one per group and prepare a presentation on this paper. The presentation will be followed by a discussion on the contents.</p> <p>Expected time budget distribution:</p> <p>attending lectures: approx. 20 hrs</p> <p>studying course notes and homework assignment: approx. 52 hrs</p> <p>Preparing presentation: approx. 40 hrs</p> |   |
| <b>Assessment</b>                      | Homework assessment and Presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Remarks</b>                         | Previous course codes were mt44060 and mt1419                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact</b>                         | <p>Tom van Terwisga: t.v.terwisga@marin.nl</p> <p>On Wednesday at TU Delft, 3ME-MTT, room 34B-1-200</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| MT44070                                | Shipping Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. E.B.H.J. van Hassel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The course offers a comprehensive insight into the transport economic aspects of a maritime transport chain. This includes:</p> <ul style="list-style-type: none"> <li>- Short history of maritime trade and transport</li> <li>- Fundamental of transport economics</li> <li>- Maritime (sub) markets</li> <li>- Risk and Risk Management in shipping</li> <li>- Ports</li> <li>- Inland navigation</li> <li>- Hinterland transportation</li> <li>- Emission reductions (legislation and current developments)</li> </ul>                                                            |   |
| <b>Study Goals</b>                     | <p>The objective of this course is to provide the students the insights into how ships are used to earn money. In order to do so, the functioning of the maritime markets needs to be understood. Also, it is the aim to explain that ships are part of a larger logistics chain and that the interaction in the port and with the other transport modes is vital. Next to that, also the working principles of a port are explained even as hinterland transport. Finally, also the current developments regarding emission reductions in the total transport chain are dealt with.</p> |   |
| <b>Education Method</b>                | Physical lectures with audiovisual presentations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Final exam in written form (75%).<br/>Group coursework (2 to 3 students): a short paper based on an assignment for which a model will be provided to the students (25%).</p>                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | Old course code: MTM313-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| MT44085                                | Ultimate Strength of Marine Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. C.L. Walters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The goal of this course is to assess the safety of a marine structure subject to the ultimate limit state. A solid understanding of the ultimate limit state and ways to model it are developed. The main focus is on buckling and fracture. Analytical and numerical methods will be applied to both failure mechanisms.</p>                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                     | <ul style="list-style-type: none"> <li>- explain which load and structural parameters influence the buckling and ultimate strength and the reason why</li> <li>- calculate the buckling and ultimate strength of a stiffened panel using hand calculations and the finite element method</li> <li>- judge modelling choices in non-linear finite element methods applied to stiffened panels</li> <li>- assess the safety of the grillage against brittle fracture</li> <li>- assess the safety of the grillage due to an accidental overload</li> </ul> |   |
| <b>Education Method</b>                | Two lectures a week, some of which will be combined by workshops to improve (FE) modelling skills.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | Two reports (made in groups of 2 persons) and individual oral review. All weigh equal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Remarks</b>                         | Old course code: MTM815-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| MT44090                                | Fluid - Structure Interaction in Marine Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. J.H. den Besten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Dr. A. Grammatikopoulos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>In order to define the topology and scantlings of body shape and structural members a fluid and structural dynamic analysis is required. The fluid dynamics analysis results include rigid body displacements and loading; input for the structural dynamics analysis, providing the response: body deflections and {strain, stress, energy}. In case the body deflections are small in comparison to the rigid body displacements, a one-way coupling (a hydro-structural analysis) is sufficient. However, if the body deflections become larger, the fluid dynamics; the loading, becomes non-negligibly affected in turn meaning a two-way coupling (a hydro-elastic analysis) is mandatory. Although the rigid body displacements hardly change, the response consequences can be significant as well as the limit state (i.e. buckling &amp; ultimate strength, fatigue &amp; fracture) effects. To be able to establish the hydro-structural and hydro-elastic response and limit state characteristics of marine (i.e. fixed and floating offshore) structures, course content includes:</p> <p>Fluid structure interaction phenomena, i.e. hydro-structural and hydro-elastic response aspects; one-way/two-way coupling formulations, for:</p> <ol style="list-style-type: none"> <li>1. continuous wave loading: <ol style="list-style-type: none"> <li>a. local, regional and global (whipping) structural response considerations</li> <li>b. frequency domain solutions</li> <li>c. testing (measurement techniques)</li> </ol> </li> <li>2. transient wave loading: impact (external: slamming, internal: sloshing) <ol style="list-style-type: none"> <li>a. Von Karman and Wagner slamming model, including inertia and free vibration stage characteristics in case of wedge and cylinder water entry as well as wave impact</li> <li>b. local, regional and global (springing) structural response considerations</li> <li>c. time domain solutions: implicit/explicit numerical integration using Newmark and 4th order Runge-Kutta formulations</li> <li>d. testing (measurement techniques)</li> </ol> </li> </ol> <p>Fluid dynamics: define and select body shape topology and scantlings using rigid body displacements and loading based on:</p> <ol style="list-style-type: none"> <li>1. hydrodynamic boundary value problem formulations</li> <li>2. local pressure- and global force/moment distributions</li> </ol> <p>Structural dynamics: define and select topology and scantlings of structural members using body deflections and stress/strain/energy response based on:</p> <ol style="list-style-type: none"> <li>1. Euler/Timoshenko beam element formulations</li> <li>2. modal decomposition (frequency; natural (vibration) and wave (flexible) mode)</li> <li>3. dry/wet characteristics</li> <li>4. PS-SB symmetry (MB,V) and anti-symmetry (MB,H, MT) components</li> </ol> <p>Hydro-structural and hydro- elastic design criteria: loading and structure natural period aspects at local, regional and global level</p> <p>Limit state consequences: buckling &amp; ultimate strength and fatigue &amp; fracture</p> <p>Scaling (for global hydro-structural and hydro-elastic response testing purposes):</p> <ol style="list-style-type: none"> <li>1. mass and stiffness properties and distributions</li> <li>2. dry/wet natural frequency and natural (vibration) and wave (flexible) mode shape characteristics</li> <li>3. backbone cross-section definition (hull girder similarity)</li> </ol> |   |
| <b>Study Goals</b>                     | <p>Theory (fundamentals):</p> <p>to identify and describe the fluid structure interaction physics and phenomena for continuous/transient wave loading and vortex induced vibrations at local/regional/global marine structural response level</p> <p>to establish hydrodynamic boundary value problem formulations as well as local pressure- and global force/moment distributions in order to define and select body shape topology and scantlings</p> <p>to establish Euler/Timoshenko beam formulations, modal decomposition, dry/wet characteristics and symmetry/anti-symmetry components in order to define and select topology and scantlings of structural members</p> <p>to model hydro-structural and hydro-elastic phenomena for continuous/transient wave loading and vortex induced vibrations mathematically based on different formulations developed over time</p> <p>Validation (experiments):</p> <p>to evaluate dedicated fluid structure interaction testing techniques</p> <p>to apply scaling laws for global hydro-structural and hydro-elastic response testing purposes w.r.t. mass and stiffness properties and distributions as well as dry/wet characteristics in order to establish backbone hull girder cross-section similarity</p> <p>to analyse and process measurement data and establish hydro-structural and hydro-elastic design criteria</p> <p>Application (engineering):</p> <p>to identify and classify if a hydro-structural or hydro-elastic structural response calculation is sufficient</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

|                         |                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Education Method</b> | to calculate the hydro-structural and hydro-elastic response for analysis and design purposes in order to establish and improve the marine structure integrity & longevity; limit state consequences                                                                                                                                                                       |
| <b>Assessment</b>       | Lectures (blended)                                                                                                                                                                                                                                                                                                                                                         |
| <b>Remarks</b>          | For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB, MAPLE and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved. |
| <b>Department</b>       | Course is programmed as a 2nd year MSc course.                                                                                                                                                                                                                                                                                                                             |
|                         | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                             |

| MT44095                                | Design of Advanced Marine Vehicles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Ir. J.L. Gelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | In this course, the following will be covered:<br>Various types of advanced ship types and their specific pros and cons<br>Physics behind the advanced vessel concepts<br>Matching materials and construction methods<br>Propulsion options                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Study Goals</b>                     | The student must be able to:<br>1. Give an overview of the characteristics of the presented advanced vessels types<br>2. Understand the physics which yield the special characteristics of specific advanced vessel types<br>3. Understand which propulsion types fit which advanced vessel type<br>4. Understand material choices and construction methods, appropriate for the various advanced vessels<br>5. Understand the design aspects and principles of the various advanced vessels types<br>6. Understand the multi-disciplinary interactions in designing advanced vessels |   |
| <b>Education Method</b>                | The theoretical part will be delivered in lectures by Jaap Gelling and some guest lecturers and will be the basis for the written exam.<br><br>Secondly, for deepening the knowledge on a specific subject, a paper has to be written in groups, based on research, design activities or literature study. The subject of the paper has to be agreed upon with the responsible teacher. This work will be executed in groups of two students.<br><br>Thirdly, a short presentation of the paper by each group                                                                         |   |
| <b>Assessment</b>                      | 1 Written exam with closed book.<br>2 Delivering of a paper based on a plan of approach to be agreed upon with the responsible teacher. The group size for this assignment will be approx 4.<br>3 10 minutes presentation of the paper<br><br>The written exam will count for 65% of the final grade, the paper and presentation together for 35%                                                                                                                                                                                                                                     |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| MT44100                         | Internal Combustion Engines A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 |
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| Responsible Instructor          | Prof.dr.ir. B.J. Boersma                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Instructor                      | R.H. van Till                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Education Period                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Start Education                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Exam Period                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Course Contents                 | PERFORMANCE<br>Introduction: Engine types and construction, historical overview and present applications.<br>Performance parameters: fuel economy, power density and emissions, basic principles and main paramaters of modern diesel engines.<br>Sizing of main dimensions and database of engines.<br>Thermodynamic analysis of cylinder process, including explanation of generalised polytropic process.<br>Performance analysis of diesel engines: important trends made clear by using a realistic Seiliger process and reasonable estimation of mechanical and heat losses.<br>TURBOCHARGING<br>Introduction: mechanical charging versus exhaust driven turbocharger, constant pressure versus pulse system, turbocharging explained in P-V and T-S diagram.<br>Gas exchange: mechanisms of air supply and air swallow capacity of 2- and 4-stroke engines, mechanisms of gas disposal, i.e. blow down, exhaust and scavenging.<br>Principles of interaction between the turbocharger and the engine: power balance of turbocharger and flow balance between engine and turbine, energy in the exhaust gas, off-design performance of a turbocharged diesel engine.<br>Enhanced turbocharging: waste gate, Variable Turbine Geometry (VTG), sequential turbocharging, two-stage turbocharging, including recent modelling developments.<br>Modelling: classification of diesel engine simulation models, physical balances, blockdiagram of engine + turbocharger.<br>NEW IN 2017: Introduction to GASENGINES (LNG as a fuel)                                          |   |
| Study Goals                     | The student must be able to:<br>1. Recognise the technical and economical importance of the diesel engine relative to other energy generating installations.<br>2. Explain the complexity and interdependency of the main performance parameters of a diesel engine and be able to calculate these parameters in several problem settings.<br>3. Formulate the limitations when determining the main dimensions of a diesel engine and be able to calculate these dimensions in several problem settings.<br>4. Analyse the thermodynamic processes in the diesel engine, both in the cylinder and in the turbocharger and be able to calculate the thermodynamical process data and resulting engine performance.<br>5. Discuss the importance of main performance parameters as well as trends and limitations from the perspective of the user<br>6. Explain the principles of a turbocharger as an example of direct waste heat usage to increase the power density of the engine.<br>7. Explain the gas exchange mechanisms and be able to contrast the differences between the 2- and 4-stroke diesel engine.<br>8. Explain the complex interaction between engine and turbocharger at part load and be able to calculate part load performance with a simplified method.<br>9. Discuss the importance, trends and limitations of turbocharging from the perspective of the user.<br>10. Identify the relevant characteristics of LNG fuelled engines (including Dual Fuel engines)<br>11. Apply the knowledge gained during the course in a practical laboratory test. |   |
| Education Method                | The student prepares the lecture by reading chapters and preparing (preferably challenging) questions. During the lectures these questions are discussed as well as questions raised by the tutor. The lecturing is focussed on subjects that are agreed to need further explanation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Assessment                      | Written exam, open book. In case of a remote/online exam, the exam will contain several fraud revention measures, like Oath of integrity, Check handwritten explanations/ answers on similarity, Oral check after exam, Several time slots for different questions, Shorter timeslot, Different versions of one question, Shuffled questions, Parametrize questions, Plagiarism check.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Remarks                         | Previous code: WB4408-A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Department                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| MT44105                                | Internal Combustion Engines B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
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| <b>Course Coordinator</b>              | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>Course Contents FUELS</p> <p>Refining process, chemical structure of hydrocarbons, types and designation of fuel products, availability.</p> <p>Fuel properties: composition (C/H ratio, S-content), density, viscosity, combustion value, ignition performance (CCAI).</p> <p>Fuel (pre)treatment and separation.</p> <p>NEW IN 2017: NEW AND RENEWABLE FUELS AND THEIR POTENTIAL OF EMISSION REDUCTION FOR INTERNAL COMBUSTION ENGINES</p> <p>COMBUSTION</p> <p>Stoichiometric number, mol and mass balance of combustion process, analysis of exhaust gas composition, practical determination of air excess ratio from emission measurements</p> <p>Properties of substances and mixtures: series expansion for specific heat, internal energy, enthalpy and entropy.</p> <p>Thermodynamics of combustion: precise definition of heat of combustion</p> <p>Chemics and physics: atomizing, ignition (conditions, delay), explosion diagram, flame shape, thermal theory (Arrhenius), reaction mechanisms, chain reactions, premixed and diffusive combustion.</p> <p>Heat release: Determination of Net and Gross Apparent Heat Release as well as Combustion Reaction Rate by reversed simulation algorithm, practical measurement (incl. assessment of Top Dead Centre), maximum pressure and temperature as a result of heat release.</p> <p>Modelling of heat release by single and double Vibe model, basic principle of single zone cylinder process simulation.</p> <p>HEAT TRANSFER</p> <p>Mean values of heat transfer between gas and wall: summation over time and location.</p> <p>Heat transfer mechanisms: conduction, convection, gas radiation and flame radiation, order of magnitude of the different contributions.</p> <p>Heat transfer coefficients: empirical methods (Nusselt, Eichelberg), methods based on dimension analysis (Sitkei, Annand, Woschni).</p> <p>Gas velocities in the cylinder: swirl, squish.</p> <p>(For information only:</p> <p>Dynamics of heat transfer: theory of instationary heat transfer in the cylinder wall: stepchange, sinusoidal and periodical input.</p> <p>Estimation of temperature amplitude in the wall.)</p> <p>EMISSIONS</p> <p>Overview: types of emission: gaseous (CO<sub>2</sub>, CO, HC, SOX, NOX) and particles (PM).</p> <p>Requirements: units and methods of conversion, measurement techniques and equipment.</p> <p>Formation: chemics, i.e. reactions, equilibrium and kinetics</p> <p>Hazards: health and environmental.</p> <p>Methods of reduction:</p> <ol style="list-style-type: none"> <li>1. Choice of fuel.</li> <li>2. Primary methods (engine optimisation): injection(timing/pressure/rate/shape), air humidity, air inlet temperature, compression ratio, air excess ratio, variable turbine geometry (VTG).</li> <li>3. More drastic primary methods (engine modifications): fuel/water emulsion, water injection, exhaust gas recirculation (EGR).</li> <li>4. Secondary methods (end of pipe solutions): selective catalytic reduction (SCR).</li> </ol> <p>Legislation: stationary installations, road traffic and shipping.</p> |   |
| <b>Study Goals</b>                     | <p>The student must be able to:</p> <ol style="list-style-type: none"> <li>1. Recognise the global problem of fossil fuels availability and the undesirable effects of combustion for the environment i.e. pollutant emissions into the air. Understand the potential of renewable fuels in this aspect.</li> <li>2. Explain the chemical and physical aspects of combustion in a diesel engine.</li> <li>3. Explain the methods to determine the heat release in a diesel engine and differentiate between levels of heat release (net, gross, reaction rate).</li> <li>4. Explain the principles used for simulation of the cylinder process in the diesel engine, in particular the modelling of the heat release.</li> <li>5. Calculate exhaust gas composition and calculate emissions into units as required by legislation.</li> <li>6. Describe aspects of the emission problem, i.e. types of emissions, formation, hazards, methods of reducing emissions and legislation.</li> <li>7. Explain the relation between the cyclic nature of heat transfer and spatial and temporal mean values.</li> <li>8. Describe mechanisms of heat transfer and apply these to the diesel engine.</li> <li>9. Explain the differential equations describing the dynamic heat transfer in the wall. Explain the solutions for step, sinusoidal and periodical.</li> <li>10. Apply and modify a model for heat release calculation.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>The student prepares the lecture by reading chapters and preparing (preferably challenging) questions. During the lectures these questions are discussed as well as questions raised by the tutor. The lecturing is focussed on subjects that are agreed to need further explanation.</p> <p>Assignment with modelling heat release.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>Written exam, open book. In case of a remote exam, the exam will contain several fraud prevention measures, like Oath of integrity, Check handwritten explanations/ answers on similarity, Oral check after exam, Several time slots for different questions, Shorter timeslot, Different versions of one question, Shuffled questions, Parametrize questions, Plagiarism check.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Remarks</b>                         | Old course code: WB4408B-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| OE44085                                | Fatigue Strength of Marine Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. J.H. den Besten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>Marine structures active in inland, coastal, offshore and deep(sea) waters are exposed to cyclic mechanical loading, forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading &amp; response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the marine structure integrity &amp; longevity in this matter, course contents include:</p> <p>Fatigue fundamentals, i.e. physics and damage process phenomena in:</p> <p>materials at defect locations (stress concentrations at micro- and meso-scale); different sources and different stages in the fatigue &amp; fracture process (crack initiation, growth and propagation, fracture).</p> <p>structures at notch locations (stress concentrations at macro-scale) as part of structural members or at the member connections; particular attention will be paid to arc-welded joints in that respect, since marine structures are predominantly assemblies consisting of plates, shells and tubular members.</p> <p>Macroscopic (notch) stress concentrations, distributions and intensities:</p> <p>as part of structural members.</p> <p>at (welded) structural member connections, consisting of a cyclic mechanical- and quasi-constant thermal residual part.</p> <p>Fatigue resistance:</p> <p>materials (plane geometries)</p> <p>structures (notched geometries: welded joints) at small-, large- and full-scale</p> <p>testing (measurement techniques)</p> <p>damage criterion life time relations in different ranges (low-, medium- and high-cycle fatigue); formulations and parameter estimates using regression analysis (least squares- and likelihood approaches).</p> <p>micro- and macro (short and long) crack growth relations.</p> <p>Fatigue assessment concepts:</p> <p>damage criteria using an intact geometry parameter (nominal stress-, structural hot spot stress- and effective notch stress/ strain concept).</p> <p>damage criteria using a cracked geometry parameter (crack tip stress intensity- and equivalent stress concept).</p> <p>Particular attention will be paid to analytical formulations and numerical approaches; concept capabilities and limitations.</p> <p>Fatigue influence factors: material and geometry (strength, surface integrity, volume, welding improvement techniques), environment (temperature, corrosion), loading &amp; response (min/max ratio, mean stress, amplitude variability (time domain analysis, cycle counting, spectral approaches, scatter diagrams, cumulative damage hypotheses), multi-axiality including proportionality (in-phase, out-of-phase).</p> <p>Fatigue reliability for (risk based) inspection and maintenance purposes: time dependent limit state functions, (joint) design variable statistics, level 1, 2 and 3 analysis, 1st and 2nd order analytical reliability methods and Monte Carlo simulation.</p> <p>Structure integrity &amp; longevity: fatigue strength calculation and life time estimation for assessment and design using different (design) philosophies, concepts, rules, standards and guidance.</p> |   |
| <b>Study Goals</b>                     | <p>Marine structures active in inland, coastal, offshore and deep(sea) waters are exposed to cyclic mechanical loading, forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading &amp; response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the marine structure integrity &amp; longevity in this matter, course objectives include:</p> <p>Theory (fundamentals):</p> <p>to identify and describe the fatigue physics and damage process phenomena in materials &amp; structures</p> <p>to establish and calculate the macroscopic fatigue characteristic structural response parameters</p> <p>to investigate fatigue resistance testing considerations and characterise damage criterion life time relations</p> <p>to model fatigue physics mathematically based on different damage criteria developed in the course of time</p> <p>to assess fatigue resistance consequences of relevant influence factors and incorporate models in fatigue damage criteria</p> <p>to examine 1st and 2nd order as well as Monte Carlo simulation based reliability methods at level 1, 2 and 3 for (risk based) inspection and maintenance purposes</p> <p>Validation (experiments):</p> <p>to evaluate dedicated fatigue testing techniques</p> <p>to analyse and process measurement data and establish fatigue resistance parameters</p> <p>Application (engineering):</p> <p>to identify and classify fatigue sensitive locations in structures</p> <p>to select an appropriate fatigue assessment concept</p> <p>to establish and improve marine structure integrity &amp; longevity, i.e. to calculate the fatigue strength and corresponding life time estimate for analysis and design purposes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Method</b>                | Lectures (blended)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Tags</b>                            | Analysis<br>Modelling<br>Structural Mechanics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| OE44150                                | Marine Hydromechanics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 |
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| <b>Responsible Instructor</b>          | Dr. H. Seyffert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | HC 4.0.0.0<br>W 2.0.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | The main topics of this course are:<br>- hydrostatics / floating stability<br>- constant potential flow<br>- constant real (viscous) flow<br>- free surface (regular & irregular) waves<br>- wave spectra<br>- wave forces on slender structures                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | After this course, students should be able to:<br>- Identify the relevant hydromechanical principles of the ocean environment for the analysis of offshore systems<br>- Formulate the hydromechanical principles of the ocean environment for the analysis of offshore systems<br>- Determine the hydromechanical loading on fixed rigid offshore systems<br>- Assess the implications of simplifications of hydromechanical principles and corresponding assumptions on the analysis of fixed rigid offshore systems |   |
| <b>Education Method</b>                | classes, practice workshops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Offshore hydromechanics, third edition, by Massie, Journée and Huijsmans<br><br>Handout for wave forces on slender structures                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | written                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |



|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

**Star Electives Ethics: Select 1 course (3 ECTS)**

| TPM002A                                | Water Ethics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Prof.dr.mr.ir. N. Doorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible for assignments</b>     | Prof.dr.mr.ir. N. Doorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. U. Pesch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | <p>Water is essential for life. While we can in principle survive weeks without food, without water, we would not survive more than a couple of days.</p> <p>The aim of this course is to provide an introduction to philosophical water ethics. It will present the main philosophical issues in the water domain in a practical way by situating these issues within real cases. Questions addressed will include:<br/>         Is it fair to drink water, knowing that the production of one cup of coffee requires 140 litres of water on average?<br/>         Should we prioritize between different water uses, like agriculture, navigation, and recreation?<br/>         Should we continue building hard flood defences at the expense of the environment?<br/>         Should local citizens be given a voice in water-related decision making?<br/>         These and other questions will be linked to ongoing discussions in the relevant other disciplines within philosophy, most notably in ethics of technology, philosophy and ethics of risks, environmental ethics, climate ethics, and global justice.<br/>         This course will provide students with an understanding of approaches in water engineering and water policy. Students will get experience with discussing and evaluating ethical considerations in water engineering and water policy. Topics covered include: water and justice, water and economic valuation, water and human rights, water and responsibility, and water and engineering.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>Describe the world's water challenges, also in view of climatic and geographic developments.</li> <li>Describe and analyse the different water uses and explain how they may lead to competing demands;</li> <li>Describe how the world's water challenges pose questions of justice and apply different concepts of justice to concrete water problems;</li> <li>Recognize and critically reflect on the methodological and ethical aspects of economic valuation in the context of water;</li> <li>Describe and apply different legal frameworks and rights concepts to the context of water;</li> <li>Describe and apply different concepts of responsibility to the context of water;</li> <li>Recognize and analyse the value-ladenness of water engineering and design;</li> <li>Describe and analyse current water governance practices and identify opportunities to include ethical considerations in water governance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>                | The structure of classes will combine lectures with seminars and debates. Dependent on the Corona restrictions, the course may take place on campus or online.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Books</b>                           | "Water ethics: An introduction", by Neelke Doorn. Publisher: Rowman & Littlefield. 2019. ISBN: 9781786609519 (Pb) / 9781786609526 (E-book).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>This course has no formative assessments or exams. It will be assessed via written assignments. The assessment consists of three parts:</p> <ul style="list-style-type: none"> <li>Case report (group assignment in groups of 2-4 students);</li> <li>Review (individual assignment);</li> <li>Reflection (individual assignment).</li> </ul> <p>Calculation of the course grade: The course grade is calculated as follows: <math>0.5 * \text{case report grade} + 0.25 * \text{review grade} + 0.25 * \text{reflection grade}</math>. The final grade will be rounded to the nearest half-integer. Students with a final grade of 6.0 or higher have passed the course.</p> <p>Validity: Grades for the different parts are valid only in the academic year in which the student participates in the tutorial sessions.</p> <p>Resit: In case of a grade lower than 6.0, students are allowed to write a new case report on a different case. This needs to be completed in the quarter following the quarter in which the tutorial sessions take place.</p> <p>Feedback: Students receive feedback on their assignments via Brightspace.</p> <p>Note: The assessment form does not depend on whether the course will be delivered online or on campus.</p>                                                                                                                                                                                                                                                                           |   |
| <b>Enrolment / Application</b>         | <p>The course TPM002A is the 3 ECTS version of the water ethics course. There is also a 5 ECTS version of the water ethics course, with course code TPM003A. Students enrolled in the 3 ECTS course follow exactly the same study activities as the students enrolled in the 3 ECTS course, except that they don't write the additional essay that students enrolled in the 5 ECTS version of the course need to write as one of their final assignments. This essay is worth 2 ECTS. All students should enrol in the 5 ECTS version of the course, also the students who do not wish to be eligible for this additional 2 ECTS. In the first week of the course, students can indicate whether they wish to follow the 3 ECTS or the 5 ECTS version of the course. There is no separate active Brightspace site for the 3 ECTS course (TPM002A).</p> <p>Please enrol for the course via the Brightspace site of TPM003A, also if you want to enrol for the 3 ECTS version of the course. Enrolment is possible until 1 November of the academic year. On that day, the total number of available places is determined. After 1 November, enrolment takes place upon approval and may be denied if there are no more places available.</p>                                                                                                                                                                                                                                                                                                |   |
| <b>Special Information</b>             | Attendance of the tutorial sessions is mandatory. Students are allowed to miss only one session and only if they have a valid reason.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Remarks</b>                         | For ethics teaching at TU Delft, check <a href="https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/">https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | Ethics<br>Sustainability<br>Water Engineering<br>Water management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Targetgroup</b>                     | All students enrolled in a MSc-program at TU Delft with an interest in water ethics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| TPM404B                                | Technology Entrepreneurship and Internationalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | F. Delgado Medina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship.</p> <p>Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.</p> |   |
| <b>Study Goals</b>                     | <p>Upon finishing the course you will be able to:</p> <p>LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms.<br/> LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas.<br/> LO 3 Identify and analysis international opportunities across different markets.<br/> LO 4 Describe and apply the born-global start-up growth process.<br/> LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process.<br/> LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time.<br/> LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies.<br/> LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.</p>                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.</p> <p>In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype).</p> <p>Book:<br/> Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted in Brightspace<br/> Slides presented in class: Will be posted in Brightspace</p>                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The assessment is as follows:</p> <ul style="list-style-type: none"> <li>- Team: Case analyses (30%) and a final report and presentation (40%)</li> <li>- Individual: theory quizzes (30%)</li> </ul> <p>All marks will be expressed on a scale of 1-10.</p> <p>In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| WM0320TU                               | Ethics and Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 |
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| <b>Module Manager</b>                  | Dr. F. Santoni De Sio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | M. Sand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>  | M. Sand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | This course is identical to the initial part of the course WM0329TU.<br>You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems. |   |
| <b>Study Goals</b>                     | After having completed the course you:<br>can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology.<br>have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds.<br>are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers.<br>are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.                                |   |
| <b>Education Method</b>                | A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | Written exam (60%), presentation and active participation during the working group sessions (40%).<br><br>In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75.<br><br>It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.                                                                                                                                                                                                                                                                                                          |   |
| <b>Enrolment / Application</b>         | Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                         | The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Category</b>                        | MSc niveau                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| WM0349WB                               | Philosophy of Engineering Science and Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. J.M. Duran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Dr. J.M. Duran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Responsible for assignments</b>     | Dr. J.M. Duran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Co-responsible for assignments</b>  | Ir. S.J. Zwart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Course contents:</p> <ol style="list-style-type: none"> <li>(1) The goals of science; the character and scope of scientific claims.</li> <li>(2) The goals of engineering design; the nature of technical artefacts; the value-neutrality of technology.</li> <li>(3) The scientific method and the validation of scientific claims.</li> <li>(4) Methods of engineering design; the character and scope of design claims; the decision-making aspect of design.</li> <li>(5) The development of science; the objectivity of science; the notion of scientific progress.</li> <li>(6) The development of technology; social determinism and technical determinism.</li> <li>(7) The place and role of values in science and in engineering design.</li> </ol> |   |
| <b>Study Goals</b>                     | This course aims first of all to support students in developing a reflective and critical attitude with regard to empirical research underlying engineering science and engineering design at an academic level. Additionally, and in support of this primary goal, it aims to make students acquainted with views on the nature and goals of science and engineering design and how in these activities facts and values both have their role to play and how they interact.                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | The course is taught in the form of seven weekly sessions of a plenary lecture of two hours, and five seminars of two hours during which students articulate and discuss their answers to a number of questions in smaller groups. Each seminar will be prepared and chaired by a team of about three students. For this task general and specific instructions, recommendations and suggestions will be available.                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | Assessment is through (1) a final individual exam, (2) participation in the seminars and (3) performance in preparing and chairing a seminar session. The exam result forms 60 percent of the final grade, assessment of general seminar participation 20 percent and assessment of the preparing and chairing role also 20 percent. In the expectation that no COVID restrictions will be in place while the course is being offered, the exam will be a closed-book on-campus exam, which may be either hand-written or digital (Möbius). Which of the two will be announced at the start of the course.                                                                                                                                                       |   |
| <b>Tags</b>                            | Artificial intelligence<br>Ethics<br>Philosophy<br>Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| WM0353TU                               | Climate Ethics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | A. Melnyk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Module Manager</b>                  | Prof.dr.ir. B. Taebe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | A. Melnyk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Responsible for assignments</b>     | A. Melnyk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Co-responsible for assignments</b>  | Prof.dr.ir. B. Taebe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/3/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | <p>Climate change is considered to be one of the most urgent problems the world is currently facing. Two types of solutions are proposed namely mitigation and adaptation. Both climate change mitigation and adaptation are heavily relying on the progress of technology. Addressing the climate problem requires, however, more than just developing and applying a certain technology. It demands considerations of whose interests and whose rights are at stake. It also requires reflection on how and "under which condition" technology can change and shape the world for the better.</p> <p>In this course we will focus on the following ethical aspects of climate change:</p> <ul style="list-style-type: none"> <li>- Past emissions and responsibility to deal with climate change</li> <li>- Implications of global warming on human safety and security</li> <li>- The distribution of burdens and benefits, emission rights and international justice</li> <li>- Future generations and intergenerational justice</li> <li>- New developments after the Paris Agreement</li> </ul> <p>The course will be offered in English and it is primarily for MSc students. Enrollment on Bright Space is sufficient.</p> |   |
| <b>Study Goals</b>                     | <ul style="list-style-type: none"> <li>- Creating awareness about climate change induced by humans</li> <li>- Insights into ethical aspects of climate change and into implications for law and policy-making</li> <li>- Being able to understand the ethical and social conditions under which technological solutions can be successful</li> <li>- Being able to understand and reflect on the international negotiations and agreements for dealing with climate change</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | Lectures and discussion sessions. Students are expected to actively participate in the class, which is why attendance is mandatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | We will study different papers and book chapters on the ethics of climate change. The required reading material along with instruction on how to acquire this material will be announced on BB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Class evaluation (1/5) and written exam (4/5). More instruction regarding the exam will be provided in class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Remarks</b>                         | For ethics teaching at TU Delft check <a href="https://www.tudelft.nl/ethics/">https://www.tudelft.nl/ethics/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| WM0375TU                        | Ethics of Technological Risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Module Manager                  | Prof.dr. S. Roeser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Instructor                      | Prof.dr. S. Roeser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Instructor                      | K. Maheshwari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Responsible for assignments     | Prof.dr. S. Roeser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Co-responsible for assignments  | Prof.dr.mr.ir. N. Doorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Education Period                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Start Education                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Exam Period                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Course Contents                 | This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations. |   |
| Study Goals                     | This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Education Method                | Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Assessment                      | On campus written exam (individual assignment), this will count for 100% of the grade for this course.<br><br>Students need to have a pass (5.75 or higher) in order to pass the course.<br><br>There is a resit of the written exam in the 4th quarter (date to be announced).<br><br>In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam.<br><br>[There is also a 5 ECTS version of this course, i.e. WM0376TU; the only difference is that for the 5 ECTS version students also write an essay in addition to the exam.]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Special Information             | Contact person for additional questions: S.Roeser@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Remarks                         | For ethics teaching at TU Delft check <a href="https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/">https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Elective                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Tags                            | Ethics<br>Philosophy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Category                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| WM0801TU                               | Introduction to Safety Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. A.P. Afghari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ir. P.H.A.J.M. van Gelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr. A.P. Afghari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-responsible for assignments</b>  | Dr. E. Papadimitriou                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/3/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Required for</b>                    | Students wishing to graduate with a final project in the area of safety supervised by the Safety Science and Security Group should have followed this course or one of the other introductory courses run by the group (WM0808TU, WM0821TU or WM0822TU)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Expected prior knowledge</b>        | Technical academic BSc and common (scientific) sense                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Summary</b>                         | This course focuses on the development of a safety-oriented pattern of thinking and a holistic approach towards systems safety and security. The tools that will be learned in this course will be helpful in recognizing, understanding, and analyzing hazards; and assessing risk in contemporary complex systems. Areas included are (1) Hazard analysis and safety assessment, (2) Human reliability assessment, and (3) Safety and security management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | Bayesian inference<br>Consequence assessment<br>Decision making under uncertainty<br>Fault & event trees<br>Human factors in safety science<br>Introduction to system safety engineering<br>Reliability block diagram<br>Risk ranking<br>Safety performance measurement<br>System safety assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                     | At the end of this course, students will be able to: <ul style="list-style-type: none"> <li>- Explain the principles of safety, security and risk management</li> <li>- Discuss the role of human factors in systems safety</li> <li>- Explain the principles of systems decomposition and causal analysis of undesired events using fault/event trees and Bayesian networks</li> <li>- Analyse data with quantitative methods for safety assessment and risk estimation</li> <li>- Apply quantitative methods on systems for risk-informed decision making</li> <li>- Perform security risk analysis using game theoretical and Bayesian analysis techniques</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Plenary lectures (mandatory)<br>Exercises (carried out during lectures, and individually or in small groups of 4-5 students)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | Clifton A. Ericson, Hazard Analysis Techniques for System Safety, John Wiley and Sons; ISBN: 978-0471720195.<br>Harold E. Ronald and Brian Moriarty. (1990). System Safety Engineering and Management, John Wiley and Sons, 2nd Edition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Prerequisites</b>                   | Technical background preferred but certainly not mandatory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>This course has only one summative assessment in the form of a final exam. There is no partial assessment. In addition, several formative assessments are conducted during the lectures including quizzes and group activities. Formative assessments are not graded.</p> <p>The final exam is individual, computer-based, closed-book and on-campus. This means that students need to be present on campus and sit behind a computer to take the exam. They are NOT allowed to access the teaching materials of the course during the exam, for instance via a USB stick on the desktop computer or via printed material. They are NOT allowed to use the internet during the exam.</p> <p>The minimum grade of the final exam in order to pass the course needs to be 5.75 or higher.</p> <p>The final grade is determined by the grade of the final exam (the final exam counts 100% for the final grade).</p> <p>The final exam is scheduled on Monday 22 January 2024.</p> <p>Students can re-sit/retake the final exam too. The re-sit exam is scheduled on Monday, 11 March 2024.</p> <p>To view the exam after publishing the grades, students must make an appointment with the main teacher of the course and meet (in person) with the teacher at a defined location.</p> |   |
| <b>Enrolment / Application</b>         | Enrol through Brightspace. Minor students are automatically enrolled into the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Remarks</b>                         | This course can be expanded to a System Safety/Reliability project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |



| WM1301TU                               | Ethics of Transportation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. F. Santoni De Sio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>  | Dr. J.B. van Grunsven                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Study Goals</b>                     | <p>After having completed the course you:</p> <ul style="list-style-type: none"> <li>can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems</li> <li>can see these problems within the broader background of moral, legal and political philosophy</li> <li>have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds</li> <li>are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain.</li> <li>are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way</li> </ul> |   |
| <b>Education Method</b>                | A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | Written exam and assessment participation in working groups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                         | For ethics teaching at TU Delft check <a href="https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/">https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

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| <b>Recommended Elective Courses</b> |
|-------------------------------------|

| AE4136-22                              | CFD 2: Discretization techniques                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M.I. Gerritsma                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Required for</b>                    | Final thesis work in structure-preserving discretization techniques                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Basic calculus and physics                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed |   |
| <b>Study Goals</b>                     | Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.                                                                                                                                                                                                                                                 |   |
| <b>Study Goals continuation</b>        | After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Computer Use</b>                    | For the final assignment a computer and Matlab or Python is required                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Relations</b>                | This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | Lecture notes                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | Final assignement and oral examination.                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Enrolment / Application</b>         | Brightspace                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Special Information</b>             | M.I.Gerritma@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Tags</b>                            | Abstract<br>Algebra<br>Calculus<br>Challenging<br>Computer Engineering<br>Fluid Mechanics<br>Mathematics<br>Matlab<br>Programming<br>Software<br>Software Engineering                                                                                                                                                                                                                                        |   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>AE4263</b>                          | <b>Modeling, Simulation and Application of Propulsion and Power Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>5</b> |
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Colonna di Paliano                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Instructor</b>                      | Dr.ir. W.P.J. Visser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Instructor</b>                      | Prof. F. Casella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Instructor</b>                      | C.M. De Servi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/x/x/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Education Period</b>                | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Expected prior knowledge</b>        | AE1240-I Thermodynamics<br>AE2230-II Propulsion and Power<br>AE1205 Programming and Scientific Computing in Python for Aerospace<br>AE4238 Aero Engine Technology<br>ME45000 Advanced Heat Transfer<br>AE2235-I Aerospace Systems & Control Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Course Contents</b>                 | <p>2nd EDUCATION PERIOD<br/>Part 1 (2 ECTS)<br/>Module 1 Introduction, Context, Foundations<br/>Module 2 Conservation equations<br/>Module 3 Modeling paradigms<br/>Module 4 Numerical methods and software<br/>Module 5 Modelica<br/>Module 6 Constitutive equations<br/>Module 7 Components and system modeling<br/>Module 8 Verification and validation</p> <p>3rd EDUCATION PERIOD<br/>Part 1 (1 ECTS)<br/>Module 9 Model-based control<br/>Take Home exam</p> <p>Part 2 (2 ECTS)<br/>Module 10 - Team Project to be chosen among<br/>a Aero engine with GSP or GTPsim<br/>b Power or propulsion system with Modelica</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Study Goals</b>                     | <p>After learning the content of the course, the student will be able to:</p> <p>Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results.</p> <p>In particular, this overarching objective can be obtained by developing the following theoretical capabilities:</p> <ol style="list-style-type: none"> <li>1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose.</li> <li>2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.).</li> <li>3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model.</li> <li>4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables).</li> <li>5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation)</li> <li>6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.</li> </ol> |          |
| <b>Education Method</b>                | <p>Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents)</p> <p>Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software.</p> <p>Students are strongly advised to make use of the BrightSpace Forum (Q&amp;A) to post questions and interact with instructors and other students.</p> <p>In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Computer Use</b>                    | <p>Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim.</p> <p>More information on these programs is provided on BrightSpace: Course Information -&gt; Resources and Tools -&gt; Open Modelica and GSP</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Literature and Study Materials</b>  | Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Books</b>                           | <p>Reference text for thermodynamics:</p> <p>W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |

**Reader  
Assessment**

Reference text for numerical methods:

P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.

The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).

**Part 1: Exercises**

The first module is tested with a Quiz.

The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded.

**Part 1: Written exam**

The take-home exam is graded and covers the material of the 9 Modules of Part 1 of the course. Duration: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace.

**Part 2: Simulation project**

Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola

Graded deliverables: technical report, model files.

A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam.

**Grading**

the final grade is calculated as

Final Grade =  $0.2 \times \text{Exercises Grade} + 0.4 \times \text{Take Home Exam Part 1 Grade} + 0.4 \times (\text{Simulation Project} + \text{Oral Exam Grade})$ .

A final grade can be obtained only if the grade of the Take Home Exam Part 1 and of Part 2 are sufficient ( $\geq 6$ ).

The Part 2 oral exam can only take place after a sufficient grade ( $\geq 6$ ) on Part 1 has been obtained.

**Resit exams**

One opportunity to resit the Take Home Exam Part 1 is provided during the second quarter the course is running in.

The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

| CS4070                                 | Multivariate Data Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 |
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| <b>Responsible Instructor</b>          | Dr. D.M.J. Tax                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Responsible Instructor</b>          | Dr. J. Söhl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 1<br>2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | Required:<br><br>* Introduction Probability Theory and Statistics: see for instance<br><br>A Modern Introduction to Probability and Statistics<br>Understanding Why and How<br>Series: Springer Texts in Statistics<br>Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E.<br>2005, XVI, 488 p. 120 illus., Hardcover<br>ISBN: 1-85233-896-2<br><br>* Basic calculus<br><br>* Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution.<br>see:<br><br>David C. Lay: Linear Algebra and Its Applications<br>ISBN-10: 0321385179 ISBN-13: 9780321385178<br>©2012 Pearson)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>PART I:<br/>Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role.</p> <p>It addresses the following subjects:</p> <ol style="list-style-type: none"> <li>1. Random variables. Matlab exercise on estimation of PDF, expected value and variance.</li> <li>2. Refresher correlation. Calculating with correlation functions.</li> <li>3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise).</li> <li>4. Random signal processing, power spectral density function, white noise.</li> <li>5. AR processes, linear prediction: theory and computer exercise.</li> <li>6. Markov chains.</li> </ol> <p>PART II:<br/>A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>PART I:</p> <ol style="list-style-type: none"> <li>1. Probability Theory <ul style="list-style-type: none"> <li>- Conditional probabilities, the law of total probability, and Bayes rule.</li> <li>- Solve probability problems that require the use of axioms of probability.</li> </ul> </li> <li>2. Definition and Description of Random Variables and Processes <p>PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements.</p> <ul style="list-style-type: none"> <li>- Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF.</li> </ul> </li> <li>3. PDF/PMF and Expected Value <p>Calculate the various forms of expected value of (combinations of) random variables and random processes</p> <ul style="list-style-type: none"> <li>- For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density.</li> <li>- Calculate the PDF, PMF, expected value and variance of a derived random variable.</li> </ul> </li> <li>4. Properties of Random Processes <ul style="list-style-type: none"> <li>- Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic.</li> <li>- Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process.</li> </ul> </li> <li>5. Large Numbers Central limit theorem, law of large numbers <ul style="list-style-type: none"> <li>- Solve problems that require the use of the central limit theorem in an engineering context</li> <li>- Explain the law of the large numbers in an engineering context.</li> </ul> </li> <li>6. Statistical Estimators <ul style="list-style-type: none"> <li>- Estimated mean, variance, and correlation function</li> <li>- Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function.</li> </ul> </li> <li>7. Application to Engineering Problems and Simulations <ul style="list-style-type: none"> <li>- Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform.</li> <li>- Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,</li> </ul> </li> </ol> |   |

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|                                         | <p>PMF, expected value, variance, autocorrelation function, autocovariance function.</p> <p>PART II:<br/>By the end of the course, you should be able to:</p> <ul style="list-style-type: none"> <li>- compute properties of frequentist and Bayesian estimators in the linear model</li> <li>- derive posterior distributions and their properties</li> <li>- apply hyperparameter optimization in Bayesian models</li> <li>- apply computational methods for posterior approximation</li> <li>- derive properties of frequentist and Bayesian classification methods</li> <li>- derive properties of MCMC methods for sampling from the posterior distributions</li> <li>- derive properties of Gaussian process regression</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Education Method</b>                 | <p>PART I:<br/>Lectures, working groups (problem solving), laboratory work (a Matlab exercise)<br/>Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Books</b>                            | <p>PART II:<br/>Classes and weekly exercises.</p> <p>PART I:<br/>R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition.</p> <p>PART II:<br/>Simon Rogers and Mark Girolami<br/>"A first course in machine learning, 2nd edition"<br/>Chapman &amp; Hall</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Assessment</b>                       | <p>It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.</p> <p>The final grade of the course consists of the following components:</p> <p>Part I (50%)<br/>Lab assignments (pass/fail)<br/>Written exam (weighting 100%)</p> <p>Part II (50%)<br/>Lab assignments (pass/fail)<br/>Written exam (weighting 100%)</p> <p>Final grade calculation: <math>0.5 * \text{Part I} + 0.5 * \text{Part II}</math><br/>A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8.</p> <p>Resit/ Repair opportunities:<br/>Part I written exam: Resit (written or oral exam, depending on the number of students)<br/>Part II written exam: Resit (written or oral exam, depending on the number of students)</p> <p>In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:<br/>Part I lab assignments: Re-submit deliverable<br/>Part II lab assignments: Re-submit deliverable</p> |
| <b>Exam Hours</b>                       | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).</p> <p>PART I:<br/>Exam of 3 hours.</p> <p>PART II:<br/>Exam of 3 hours.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Permitted Materials during Tests</b> | <p>PART I:<br/>Self made notes on a two-sided written A4 sheet.<br/>Calculator.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Remarks</b>                          | <p>PART II: none</p> <p>PART II:<br/>This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| ME44125                                | Reliability and Maintenance of Transport Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | X. Jiang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior knowledge</b>        | Technical academic BSc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>Transport equipment is in general a mechanical system, growing in size, capacity and complexity in order to accommodate continuous development of transport. It is essential to ensure sufficient safety, serviceability and durability of the equipment at the reduced cost in order to maintain the effectiveness of the transport system. This belongs to the research context of system reliability. Typically, a transport equipment consists of multiple subsystems or components with different function, and they would fail in different failure modes with different mechanism behind. Contributed by various non-destructive techniques, online sensor system and data analytics, potential failure of the equipment (fault, defect, damage) could be detected, predicted and mitigated in such a way that the occurrence of a catastrophic accident would be prevented, although with some extent of uncertainty caused by, i.e., the sensitivity of NDTs &amp; sensors, environmental noise, etc. Upon the detection of a fault(or damage), a follow-up maintenance strategy shall be made in order to determine whether, when, where and how to repair/maintain the equipment, constrained by available budget, facilities and resources, etc.</p> <p>The overall objective of the course is to prepare engineering graduates with an in-depth knowledge of system reliability, maintainability and availability. A series of modules will be given covering the fundamental theory embedded in the reliability and maintenance of transport equipment. Application of the theory in practice will be exemplified and practiced through in-class exercises and discussion, weekly assignments and final written report. One or two guest lectures will be introduced to demonstrate the implication of system reliability in the operation and maintenance of the transport equipment and the infrastructure.</p> |   |
| <b>Study Goals</b>                     | <p>After successfully completing the course, students will be able to:</p> <ul style="list-style-type: none"> <li>-Define system Reliability, Maintainability and Availability(RAM)and describe associated basic measures using statistic method.</li> <li>-Carry out system reliability analysis to identify critical component / subsystem using statistic models, like FMECA, FTA, ETA and Markov process among others.</li> <li>-Apply the acquired data to predict Remaining Useful Life (RUL) using statistic, data driven or hybrid models.</li> <li>-Analyze common maintenance strategies at both component and system levels based on statistic, data driven or hybrid models.</li> <li>-Develop and reflect a maintenance strategy for a simplified transport equipment/system based on statistic, data driven or hybrid models.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Method</b>                | Lectures and supplementary material provided during lectures, in class exercises and discussions, weekly assignment, group work on final written report and presentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based on 30% of the grade for weekly assignments plus 70% of the grade for the final report and associated presentation(50/50). Four to five students are anticipated to work as a group on the final report. Weekly assignments are compulsory and they can be done either individually or as a group work (the same group on the final report).</p> <p>The grade for assignments will be the average grade of all weekly assignments. To pass the course, the grade for each weekly assignment is not allowed to be lower than a 5.0.</p> <p>The final presentation will take place on campus. In case of unforeseen circumstances or measures resulting from i.e., COVID-19 etc., the final presentation will then be given online. Detail information will be announced on Brightspace in due time.</p> <p>The final grade must be a 5.75 or higher in order to pass the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Remarks</b>                         | <p>Reference books:</p> <ol style="list-style-type: none"> <li>1. Marvin Rausand, Arnljot Høyland, System Reliability Theory: Models, Statistical Methods, and Applications.</li> <li>2. Kumar, U. D., Crocker, J., Knezevic, J., &amp; El-Haram (2000), M. Reliability, Maintainability and Logistic Support - A Lifecycle Approach.</li> <li>3. Charles E. Ebeling (2010), An Introduction to Reliability and Maintainability Engineering.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |



| ME44210                                 | Drive & Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 |
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| <b>Responsible Instructor</b>           | Dr.ir. H. Polinder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/0/0/0 (1st part Q1 2x2 hours, 2nd part Q1 1x2 hours)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>         | This course is designed for students without a BSc in Electrical Engineering. Proper knowledge of mathematics (differential equations, complex numbers) and mechanics is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                  | <ul style="list-style-type: none"> <li>- Introduction to drive and energy systems</li> <li>- Electricity and electrical circuits</li> <li>- Magnetism and magnetic circuits</li> <li>- Power electronics</li> <li>- DC machines</li> <li>- AC voltages, currents, powers and phasors</li> <li>- Transformers</li> <li>- Synchronous machines (including PM machines)</li> <li>- Induction machines</li> <li>- Choices in electrical drive and energy systems</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                      | <p>After following this course, students should be able to</p> <ul style="list-style-type: none"> <li>- Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws.</li> <li>- Sketch phasor diagrams with voltage and current phasors.</li> <li>- Calculate active, reactive and apparent power in AC circuits.</li> <li>- Calculate energy stored in capacitors and inductors.</li> <li>- Describe the most important power semiconductors (diodes, IGBTs, MOSFETs) and their switching behaviour.</li> <li>- Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs).</li> <li>- Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation).</li> <li>- Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode.</li> <li>- Calculate induced voltage using Faradays law.</li> <li>- Calculate electromagnetic forces and torques using Lorentz law or the power balance.</li> <li>- Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines.</li> <li>- Derive performance characteristics based on these equivalent circuits (torque-speed characteristics).</li> <li>- Describe advantages and disadvantages of electrification of drives.</li> <li>- Select suitable machine and drives for different applications.</li> </ul> |   |
| <b>Education Method</b>                 | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Books</b>                            | T. Wildi, "Electrical Machines, Drives and Power Systems"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Prerequisites</b>                    | Mathematics, Mechanics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | <p>The method of assessment will be: written exam, closed book, partly multiple choice. An equations form will be handed out with the exam.</p> <p>If on campus assessment is not possible (for example, because of measures resulting from COVID-19) the assessment will be as follows: written exam, open book, online, partly multiple choice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Permitted Materials during Tests</b> | An equations form will be handed out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Department</b>                       | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact</b>                          | h.polinder@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| ME44300                                | Multi-Machine Coordination for Logistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 |
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| <b>Responsible Instructor</b>          | Dr. V. Reppa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Prof.dr. R.R. Negenborn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In this course students will get familiar with automatic control techniques and their benefits and applications for real-time coordination of actions of large-scale transport systems. Theoretical concepts mostly related to model predictive control are discussed, as well as their application to real-world transport systems. Illustrative examples of large-scale systems include among others intermodal / synchromodal (i.e. road, waterborne, rail) transport networks. The operation of these networks relies on the interoperability of various machines such as cargo ships, trains, road tracks, automated guided vehicles, quay/yard cranes and many more.                                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>Students are able to:</p> <p>LO1. apply system theory to describe large-scale machines and transport systems</p> <p>LO2. explain the design of an automatically controlled system and of the automatic coordination of several controlled systems</p> <p>LO3. explain the design of control architectures for large-scale interconnected machines and transport systems (centralized, distributed, single-agent, multi-agent, single-level, multi-level)</p> <p>LO4. evaluate the pros and cons of different control architectures and their impact on multi-machine coordination for transport</p> <p>LO5. design an optimal control strategy for a single machine used in the transport network (truck, ship, crane, etc)</p> <p>LO6. develop a multi-agent framework based on model predictive control for real-time logistics of large-scale transport systems</p> |   |
| <b>Education Method</b>                | Study material will be provided to support the lectures during which new concepts will be discussed and explained. Small groups of students will be created to conduct a project that will help them to develop deeper understanding of the theoretical (optimal) control and coordination concepts applied to a specific machine that is part of a larger transport network. Throughout the course illustrative examples will be used for showing the application of new concepts. Feedback sessions will be organized to discuss questions about the project.                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Relations</b>                | The course is related to the exit qualifications 1.b, 1.c, 3.b, 3.c and 7.a of the track MME of the Master Programme Mechanical Engineering, and to the specialization Engineering of the Master Programme Transport, Infrastructure and Logistics. Prior knowledge in automation of transport systems, dynamical systems, optimization and simulation is recommended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>We will discuss chapters from the book "Intelligent Infrastructures", Negenborn et al., Springer, Dordrecht, The Netherlands, 2010. ISBN: 978-90-481-3597-4.</p> <p>Additional handouts and references to relevant scientific literature and media will also be provided.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | Assessment will be based on a written group project report (50%) and on a written individual (closed-book) exam (50%). In principle, the written exam will take place on campus. Resit options are offered for both the project and the exam. Note that the grade of the exam and the grade of the project are only valid for the academic year 2023-2024.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Information &amp; Communication</p> <p>Modelling</p> <p>Optimalisation</p> <p>Project</p> <p>Transport &amp; Logistics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact</b>                         | Dr. Vasso Reppa (v.reppa@tudelft.nl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| ME45001                                 | Advanced Heat Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
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| <b>Responsible Instructor</b>           | Dr.ir. J.W.R. Peeters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Dr. R. Delfos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>         | A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena.<br>Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | <p>In this course the concepts &amp; mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures.</p> <p>During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems.</p> <p>Subjects in order of appearance:</p> <ul style="list-style-type: none"> <li>- A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy.</li> <li>- A refresher on Conduction, Convection and Radiation.</li> <li>- Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms.</li> <li>- Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques &amp; numerical techniques; relaxation.</li> <li>- Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number &amp; Stefan condition.</li> <li>- Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques &amp; numerical techniques; stability criteria.</li> <li>- Forced &amp; Free convection: Nusselt, Stanton, Prandlt &amp; Peclet numbers; Analysis &amp; the physics behind empirical correlations. The role of boundary conditions.</li> <li>- Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.</li> </ul> |   |
| <b>Study Goals</b>                      | <p>More specifically: The student is able to</p> <ol style="list-style-type: none"> <li>1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner.</li> <li>2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches.</li> <li>3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction.</li> <li>4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions.</li> <li>5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries.</li> <li>6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations.</li> <li>7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications.</li> </ol> <p>Indicate implications when more detailed distributions of convective heat transfer are involved.</p>                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures (2x2 hours per week) including experiments &amp; computer demonstrations.</li> <li>- After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided.</li> <li>- Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment.</li> <li>- Total number of contact hours: 6 hours per week.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Computer Use</b>                     | Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>   | <p>Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used:<br/>A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'.</p> <p>The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical.</p> <p>Alternative books on heat transfer from your Bachelor's course (including Cengel &amp; Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard.</p> <p>Lecture slides, short videos, Matlab scripts &amp; other study materials required will be provided via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                       | The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Permitted Materials during Tests</b> | Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted.<br>One (1) equation sheet of 2 sides A4 can be taken and/or is provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Department</b>                       | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| ME45155                                | Computational Fluid Dynamics for Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. M.J.B.M. Pourquie                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Prof.dr. R. Pecnik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. P. Simões Costa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>        | WB2542 (thermofluids), ME45042 or ME45043 (advanced fluid dynamics), WBMT2049 T2 and T3 or similar (basic numerical analysis), some elementary programming skill (python, matlab or any other)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>This is a course on computational fluid dynamics related to process and energy systems. Some elementary knowledge of numerical methods (discretisation of a derivative, numerical solution of ordinary differential equations) is assumed. See expected prior knowledge.</p> <p>In general the course will cover:</p> <ol style="list-style-type: none"> <li>(1) general conservation equations for mass, heat and momentum transport (Navier-Stokes equations)</li> <li>(2) numerical solution of multi-dimensional, stationary and time-dependent problems as encountered in low-speed CFD applications; <ol style="list-style-type: none"> <li>(2.1) numerical schemes for temporal discretization;</li> <li>(2.2) numerical schemes for spatial discretization;</li> <li>(2.3) imposition boundary conditions</li> </ol> </li> <li>(3) turbulence modeling</li> <li>(4) wave equation</li> <li>(5) numerical methods for high-speed (compressible) flows</li> <li>(6) high performance computing</li> </ol>                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>After learning the content of the course the student will have the following capabilities:</p> <ol style="list-style-type: none"> <li>(1) Describe the two most popular methods in commercial CFD, finite differences and finite volumes</li> <li>(2) solve simple demonstrative problems in fluid flow and heat transfer by programming them in python or matlab, using finite differences and finite volumes</li> <li>(3) recognize the effects of numerical methods on the solution, such as numerical diffusion and numerical dispersion and to explain how to make these effects smaller</li> <li>(4) recognize numerical instability, to list several ways to avoid it and to analyze stability of simple methods analytically</li> <li>(5) solve fluid flow and heat transfer problems with the commercial CFD package Fluent or the open source CFD package OpenFOAM, which includes the following: set up a calculation, generate a simple grid, set appropriate boundary conditions, choose suitable discretisation, interpret and validate the results. Alternatively, in some cases, instead of using a commercial package, students can decide to write and validate their own code which they will use for calculating a fluid mechanics problem.</li> </ol> |   |
| <b>Education Method</b>                | Lectures, practical exercises (2x2 hours per week)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | <p>Course material:</p> <ol style="list-style-type: none"> <li>(1) Sheets/handouts</li> <li>(2) J.H. Ferziger and M. Peric, Computational methods for Fluid Dynamics, Springer Verlag.</li> <li>(3) P. Moin, Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2001.</li> </ol> <p>References from literature:</p> <ol style="list-style-type: none"> <li>(1) C. Hirsch, Numerical computation of internal and external flows, Volume I Fundamentals of numerical discretization, Volume II Computational methods for inviscid and viscous flows, Chicester, Wiley &amp; Sons, 1988, 1990</li> <li>(2) C.A.J. Fletcher, Computational techniques for Fluid Dynamics, Volume I Fundamental and general techniques, Volume II Specific techniques for different flow categories, Berlin, Springer, 2-nd ed. 1991.</li> <li>(3) H.K. Versteegh, W. Malalasekara, An introduction of computational fluid dynamics. The finite volume method. Second edition. Pearson Education.</li> </ol>                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | Written exam + assignment with report; written exam may be online type depending on circumstances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ME45160                                | Advanced Applied Thermodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. B.P. Tighe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. O. Moulτος                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Required for</b>                    | EFPT track                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Thermodynamics at the level of a BSc in Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <ul style="list-style-type: none"> <li>- Review of fundamentals of thermodynamics, including thermodynamic potentials, Maxwell relations, and the 1st and 2nd Law</li> <li>- Equilibrium in one-component systems: equations of state, chemical potential, fugacity, Clausius-Clapeyron, etc.</li> <li>- Mixtures: ideal gas mixtures, partial molar quantities, Gibbs-Duhem, vapor- liquid equilibrium, activity coefficient models, and osmosis</li> <li>- Model equations of state</li> <li>- First introduction to statistical mechanics</li> </ul> |   |
| <b>Study Goals</b>                     | After this course, you will be able to calculate phase and material properties in thermodynamic equilibrium for multi-phase and/or multi-component mixtures under ideal and non-ideal process conditions. You will be able to use equations of state and activity coefficient models, and to calibrate and validate those models using thermodynamic data.                                                                                                                                                                                              |   |
| <b>Education Method</b>                | Lectures, optional homework, review session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Computer Use</b>                    | homework assignments require scientific computing tools such as python, Matlab (or comparable) and basic programming methods such as for loops                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Books</b>                           | Holyst Poniewierski                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | Written final exam; open book                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| ME45170                                | Turbomachinery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. R. Pecnik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Responsible Instructor</b>          | Prof.dr.ir. S.A. Klein                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>        | Thermodynamics, Fluid dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>The course consists of the following parts:</p> <ul style="list-style-type: none"> <li>- Introduction and recap of the required thermodynamics. This class will give an overall overview of the relevance of turbomachinery, the general basics and the history. Furthermore the basic thermodynamics will be repeated (1 class)</li> <li>- Turbomachinery: basic theory and the application to the design of turbomachinery (Conservation equations and velocity triangles, turbines, compressors and pumps, 3D flow, radial machinery) (5 classes)</li> <li>- Application of compressor, pump and turbine maps. Limitations in the operation (stall, surge, NPSH etc) (1 class)</li> <li>- Gas turbine: components (turbine, compressor, combustor), gas turbine cycles, performance calculation,</li> </ul>                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>The student is able to describe the fundamental aspects of the working principle of turbomachinery.</p> <p>More specifically, the student must be able to:</p> <ol style="list-style-type: none"> <li>1. reproduce and apply thermodynamic principles of turbomachinery</li> <li>2. describe the basic fluid dynamics of turbomachinery and the way work is transferred between the working fluid and the turbomachinery, including the impact on performance and limitations of operations</li> <li>3. apply compressor, pump and turbine curves in practical applications</li> <li>4. describe different gas turbine cycles</li> <li>5. describe the influence of different losses on the performance (power and efficiency) of gas turbines</li> <li>6. synthesize different gas turbine cycles with respect to their application.</li> <li>7. describe the principles of combustion as applied in gas turbines with respect to functionality and emissions</li> <li>8. describe the part load characteristics of gas turbines and its relation to the functional characteristics of gas turbine components</li> </ol> |   |
| <b>Education Method</b>                | Lectures and homework assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>Closed book exam (60%) and home work assignments (40%).</p> <p>All homework assignments must be submitted.</p> <p>The grade for the homework assignments will only be included in the final grade if the exam grade is above 5.5/10</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Tags</b>                            | <p>Energy</p> <p>Energy &amp; Industry</p> <p>Fluid Mechanics</p> <p>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| ME46055                                | Engineering Dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. F. Alijani                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | R.A. Norte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Required for</b>                    | Multibody Dynamics A(ME41050), Multibody Dynamics B (ME41050), Nonlinear mechanics(ME46000), Nonlinear dynamics (ME46072).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The dynamic behavior of structures (and systems in general) plays an essential role in engineering mechanics and in particular in the design of controllers. In this master course, we will discuss how the equations describing the dynamical behavior of a structure and of a mechatronical system can be set up. Fundamental concepts in dynamics such as equilibrium, stability, linearization and vibration modes are discussed. If time permits, also an introduction to numerical integration techniques is proposed.</p> <p>The course will discuss the following topics:</p> <ul style="list-style-type: none"> <li>--Review of the virtual work principle and Lagrange equations</li> <li>--Linearization around an equilibrium position: vibrations</li> <li>--Free vibration modes and modal superposition</li> <li>--Forced harmonic response of nondamped and damped structures</li> <li>-- Introduction to time integration techniques</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>The student is able to select different ways of setting up the dynamic equations of mechanical systems, to perform an analysis of the system in terms of linear stability and vibration modes and to properly use mode superposition techniques for computing transient and harmonic responses.</p> <p>More specifically, the student must be able to :</p> <ol style="list-style-type: none"> <li>1. Explain the dynamic principle of virtual work and Lagrange equations, and discuss their relation to the basic Newton laws.</li> <li>2. Describe the concept of kinematic constraints and identify a proper set of degrees of freedom to describe a dynamic system.</li> <li>3. Use Lagrange equations via employing the minimum set of degrees of freedom to find the governing equations of dynamic systems, and construct these equations for systems with kinematic constraints.</li> <li>4. Use Hamiltons principle to find the governing equations of motion of dynamics systems.</li> <li>5. Find equilibrium positions and construct the linearized equations of motion by using different contributions of the kinetic and potential energies.</li> <li>6. Analyze the linear stability of dynamic systems according to their state space formulation.</li> <li>7. Explain and use the concept of free vibration modes and frequencies for multi degree of freedom systems.</li> <li>8. Apply the orthogonality properties of modes to describe the forced response of damped and undamped systems.</li> </ol> |   |
| <b>Education Method</b>                | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Course material:</p> <p>Lecture notes (available through brightspace)</p> <p>References from literature:</p> <ol style="list-style-type: none"> <li>1. M. G  radin and D. Rixen. Mechanical Vibrations. Theory and Application to Structural Dynamics. Wiley &amp; Sons, 2nd edition, 1997.</li> <li>2. J. Ginsberg. Engineering Dynamics. Cambridge University Press, 2008.</li> <li>3. D.J. Inman, Engineering Vibration. PrenticeHall, 1996.</li> <li>4. L. Meirovitch. Principles and Techniques of Vibrations. PrenticeHall, 1997.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | written exam + an assignment or a quiz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | <p>In case an assignment is given- The assignment will involve the use of a Matlab-like software</p> <p>2. Old course code: WB1418-07</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Department</b>                      | 3mE Department Precision & Microsystems Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| ME46060                                | Engineering Optimisation: Concepts and Applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M. Langelaar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ir. L.F.P. Noel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>        | Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | Formulation of optimization problems<br>Typical characteristics of optimization problems<br>Minimization without constraints<br>Constrained minimization<br>Direct and gradient-based optimization algorithms<br>Approximation concepts<br>Sensitivity analysis<br>Topology optimization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures.</p> <p>More specifically, you will be able to:</p> <ol style="list-style-type: none"> <li>1. formulate an optimization model for various design problems</li> <li>2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity</li> <li>3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity</li> <li>4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle</li> <li>5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods</li> <li>6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems</li> <li>7. describe the complications associated with the use of computational models in optimization</li> <li>8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models</li> <li>9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods</li> <li>10. identify a suitable optimization algorithm given a certain optimization problem</li> <li>11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox</li> <li>12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations</li> <li>13. describe the basic concepts used in structural topology optimization.</li> </ol> |   |
| <b>Education Method</b>                | <p>Lectures (2x2 hours per week), exercises, online tests.</p> <p>Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Books</b>                           | Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course.</p> <p>Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs.</p> <p>To pass the course, the final grade must be a 5.75 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest half/full mark (.5/.0). The maximum bonus score is 0.8. The bonus only applies to reports with grade <math>\geq 5.5</math> and submitted before the first report deadline.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | Former course code: WB1440                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Percentage of Design</b>            | 80%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Design Content</b>                  | The course is focusing on design optimization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Department</b>                      | 3mE Department Precision & Microsystems Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact</b>                         | Dr.ir. Matthijs Langelaar<br>Email: m.langelaar@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| MOT122A                                | Digital Business Process Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5 |
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| <b>Module Manager</b>                  | Prof.dr.ir. M.F.W.H.A. Janssen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. J. Ubacht                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr. F. Kleiman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. Y. Ding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-responsible for assignments</b>  | Dr. Y. Ding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Required for</b>                    | This course is part of the MOT Master programs, but can be used as an elective. MOT students have priority when there are too many enrollments. Bachelor students are excluded from participation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Business processes are at the core of high-tech organizations and consist of a large number of activities aimed at creating value for customers. These processes can be supported or executed by humans or by digital technology. In the digital age business processes are supported by, enabled and changed by technologies like the Internet of Things, Artificial Intelligence, Blockchain, Robotics and so on. Business processes are situated at the point where the business side meets the IT side and where technology should fulfill customer needs. If business processes fail or do not adapt to the needs of customers the whole business might fail. Business Process Management (BPM) is necessary for ensuring that processes are operating in concert and that these processes are adaptable to changes in the overall environment of an organization.</p> <p>In this course the relationship between strategy, technology and business processes will be illustrated, explained and discussed. Methods and tools for business process analysis and improvement will be presented, including a number of statistical and architecture design methods. New solutions will be architected for high-tech organizations. The extent to which the nature of business processes is affected by contemporary technology constitutes the core of this course. In particular to analyze technologies and their impact on business process management and to architect new solutions in the organizational context of the firm.</p> <p>Topics</p> <ul style="list-style-type: none"> <li>Aligning strategy and processes, value perspective, supply chains</li> <li>Workflow, resources, BPM elements and control</li> <li>Business process modeling (concepts, BPMN) + practical work</li> <li>Process measurement, Balanced Score Card, Activity-based Costing (ABC) + practical work</li> <li>Business process improvement strategies and approaches, path dependencies, capabilities, resource-based view, business process maturity</li> <li>Improvement methodologies (LSS, lean, six sigma, theory of constraints, technology ..)</li> <li>Statistical analyses and improvement + practical work</li> <li>Compliance by design, build-in-controls in processes and technology</li> <li>Business process automation and technology (BPM, ERP, workflow, SOA,...)</li> <li>Business processes automation (case-based, process -based, use-based)</li> <li>Agility and adaptability of BPM systems and the users, business process and agile enterprises</li> <li>AI, Business and knowledge rules</li> <li>Internet of Things, data-driven organizations</li> </ul> |   |
| <b>Study Goals</b>                     | <p>The focus is on realizing business processes enabled by digital technology for high tech organizations. By the end of the course, you should be able to improve business processes using technology to realize organizational strategies.</p> <p>Learning objectives:</p> <ol style="list-style-type: none"> <li>1. Analyze the relationship between business processes, strategy and technology</li> <li>2. Model a business process using BPMN</li> <li>3. Employ improvement methodologies to the selected business process</li> <li>4. Assess if new technologies improves the chosen business process</li> <li>5. Propose a new business process including technological and process improvements</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>-lectures</li> <li>-business process management game</li> <li>-practical work</li> <li>-assignments</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Computer Use</b>                    | <p>Planning tool. There are suitable planning tools like Trello and Asana</p> <p>BPMN modelling tool. There are several modelling tools which are appropriated like Eunomia process builder (<a href="http://www.eunomia-process.com/">http://www.eunomia-process.com/</a>), Bizagi (<a href="http://www.bizagi.com/">http://www.bizagi.com/</a>) or LucidChart (<a href="https://www.lucidchart.com/">https://www.lucidchart.com/</a>)</p> <p>Activity Based Costing (ABS) calculations and simulations using a spreadsheets, like MS Excel.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | <p>List of articles (containing classical papers and showing the state-of-the-art)</p> <p>Reader (containing an overview of main elements and explaining some methods)</p> <p>Exercises + answers (for practicing)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>Students will collaborate in groups of 4-7 students to make an assignment which should contain 1) the analysis 2) improvement and digitalization of a business process and 3) implementation aspects. Students are free to select a process that is critical or a company and can be improved by ICT. The assignment consists of three parts</p> <ol style="list-style-type: none"> <li>1. Analysis: Students select a company, analyze their strategy and select a critical business process. The existing process should be modeled (in BPMN) and the resulting product, stakeholders roles, customer needs and technology used should be analyzed.</li> <li>2. Improvement: Based on the improvement methods and instruments a proposal for business process improvements should be made.</li> <li>3. Realization: how the proposed business process should be realized by means of digital technology.</li> </ol> <p>The assignment should be handed in as a report. The report should adhere to the APA reference standards. The report cannot be updated for a higher grade after having handed in the report.</p> <p>The students are expected to prepare two presentations and some of the groups will be asked to present these during the lectures. Based on these presentations feedback will be given, which are relevant for all groups. Presentations will be assessed using a pass or fail. The report will be graded; this grade will reflect the final grade.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | Business<br>Design<br>Information & Communication<br>Software<br>Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |



| SC42001                                | Control System Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. A.J.J. van den Boom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation. Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design. Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance.</p> <p>Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>By taking this course, the student</p> <ul style="list-style-type: none"> <li>- will be able to master the introduced theoretical concepts in systems theory and feedback control design</li> <li>- will be able to practically apply these concepts to design projects and tasks</li> <li>- and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.</li> <li>- will be able to translate a predictive control problem into a standards setting and solve the predictive control problem.</li> </ul> <p>More specifically, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Translate differential equation models into state-space and transfer function descriptions</li> <li>- Rationalize differences between state-space and transfer function approaches</li> <li>- Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties</li> <li>- Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state</li> <li>- Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control</li> <li>- Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers</li> <li>- Formulate the separation principle and employ it for the design of output feedback</li> <li>- Build reference models and achieve zero steady-state error using integral control.</li> </ul> <ul style="list-style-type: none"> <li>- set up a predictive control problem.</li> <li>- Solve the standard predictive control problem;</li> <li>- Master the main analytical details in stability proofs of predictive control schemes;</li> </ul> |   |
| <b>Education Method</b>                | Lectures 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Textbook (its use is strongly recommended):</p> <p>K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009</p> <p>Available online for download:<br/> <a href="http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf">http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf</a></p> <p>The link for downloading the lecture notes for model predictive control will be available in the first quarter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>Written examination. Due to COVID-19 the exam may be an online exam.</p> <p>Disclaimer: information may change depending on the developments around the coronavirus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Department</b>                      | 3mE Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SEN1221                         | Statistical Analysis of Choice Behaviour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
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| Module Manager                  | Dr. E.J.E. Molin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Instructor                      | Dr. E.J.E. Molin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Instructor                      | Ir. S. van Cranenburgh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Responsible for assignments     | Dr. E.J.E. Molin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Co-responsible for assignments  | Ir. S. van Cranenburgh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Contact Hours / Week<br>x/x/x/x | 0/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Course Contents                 | <p>If one wants to design effective (transport) systems, one needs to understand the behavior of -future- users of these systems, e.g. the market share of a new public transport service. If one wants to design effective (transport) policies, one needs to know how people will respond to these policies, e.g. impact of a tax incentive on electric vehicle sales. This course sets out to provide students with in-depth knowledge and hands-on experience with behavioral theories, mathematical models and statistical methods that have proved very effective in understanding and predicting human choice behaviour.</p> <p>Importantly, this course takes a quantitative (i.e., mathematical and statistical) perspective on choice behaviour; this is quite different from the usual perspective in behavioral science-courses, which tends to be more qualitative. More specifically, we formalize behavioral theories into mathematical models, which we then estimate on empirical datasets using cutting edge statistical methods. In combination, the estimated econometric 'choice models' support rigorous analysis and forecasts of quantitative phenomena, thereby playing a crucial role in private and public sector policy development.</p> <p>The theories, models and methods that are taught in this course are applicable -and have been used- throughout a wide variety of domains, including but not limited to Transportation, Health, Marketing, Environment &amp; Energy, and Political sciences. As such, our course optimally prepares students for a wide range of (quantitative, empirical) topics for their graduation projects.</p> <p>The course will cover both canonical (and Nobel Prize winning) methods which are based on the notion of rational decision making, as well as more recently proposed ones which are based on notions of bounded rationality (economic psychology) and moral decision-making. Both lecturers are highly active in the international research community, and their work forms part of the academic state of the art.</p> |   |
| Study Goals                     | <p>Students will acquire an in-depth knowledge concerning the below mentioned theories, models and methods, as well as the skills to use them in their own research. Finally, they will acquire an understanding of the usefulness and limitations of these theories, models and methods; and an understanding of how they can and should be used in combination.</p> <p>Knowledge transfer will take the form of a series of lectures, whereas skills will be (further) developed in tailor made computer exercises. Specifically, during the computer exercises, students will learn how to design a statistically optimal choice experiment, and how to analyze data obtained from such experiments. In addition, they will learn how to interpret the outcomes of such analyses and how to use them to evaluate and design policies.</p> <ul style="list-style-type: none"><li>- Choice behaviour modelling methods (van Cranenburgh)<ul style="list-style-type: none"><li>o Discrete choice theory and the Logit model</li><li>o Advanced (Mixed) Logit models</li><li>o Choice models for bounded rationality and morality</li></ul></li><li>- Stated Choice data collection methods (Molin)<ul style="list-style-type: none"><li>o Data-collection paradigms</li><li>o Stated choice experiments</li><li>o Efficient design approaches.</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Education Method                | <p>Each method will we covered by means of:</p> <ul style="list-style-type: none"><li>o Lecture series</li><li>o Exercises, in which concepts taught during the lectures will be used in hands on exercises.</li></ul> <p>In case covid-regulations require this, a hybrid form of teaching will be adopted, where on campus lectures are livestreamed and computer exercises will be discussed online.</p> <p>IMPORTANT INFORMATION ON KNOWLEDGE REQUIREMENTS:</p> <p>We expect you to have substantial background in mathematics (calculus and statistics), including knowledge of mathematical modeling, and of statistical metrics and testing procedures such as variance, correlation, normal distribution, standard errors, t-test, and statistical testing of hypotheses (including the use of p-values). We also expect you to have knowledge of survey data collection concepts and methods (sample, population, sampling procedures, questionnaires). Note that students with insufficient background in mathematics and statistics will struggle to keep up with the pace of the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Assessment                      | <p>The final grade is determined by two partial exams which are held during the course and by a regular exam (in the exam week). The partial exams focus primarily on skills and practices, i.e. the analyses, modeling, and experimental design (which are practiced in the computer practice sessions). In contrast, the regular exam focuses mainly on theory and conceptual thinking. Note, however, that both exam types pose questions on theory and practical skills. Furthermore, Mobius or ANS examination software is used for both exams. Exams consist of multiple choice, true/false, and numerical questions and, to a lesser extent, open-ended (essay) questions.</p> <p>Since no re-sit will be offered for the partial exams, the way in which the final grade is determined depends on the period the exam is taken:</p> <p>regular exam period: Final grade = 7.5% * grade partial exam 1 + 7.5% * grade partial exam 2 + 85% * grade regular exam</p> <p>re-sit exam period: Final grade = 100% * grade re-sit exam</p> <p>Thus, the grades of the partial exams are only used in combination with the grade of the regular exam period. After this, the grades of the partial exams lose their validity. Also, the grades of the partial exams are not transferred to the next year and are not registered in Osiris.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Elective                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Tags                            | Analysis<br>Calculus<br>Economics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

Involved  
Modelling  
Research Methods  
Stochastics  
Transport & Logistics

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

|                    |
|--------------------|
| <b>Second Year</b> |
|--------------------|

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

|                                                   |
|---------------------------------------------------|
| <b>MT Graduation Project Obligatory (45 ECTS)</b> |
|---------------------------------------------------|

| MT54010-20                             | MT Literature Research                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Education Period</b>                | None (Self Study)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Course Contents</b>                 | <p>This is part 1/2 of your master graduation project; it consists of the problem analysis and literature research. This part is assessed on basis of your literature study report, which includes a plan of approach for the thesis part (second part of the graduation project), a.k.a. MT54035.</p> <p>The text below is the same as for MT54035.</p> <p>Before you can start your master thesis project you will need to hand in a signed MSc Thesis form (form 4) with the Director of Studies (P. de Vos). This form is available from the Master Marine Technology page on Brightspace. The form registers amongst others the working title of your thesis, your supervisor, a short description of your project, the start date and provisional graduation date. The form is signed by the Director of Studies only when your study progress as registered in OSIRIS is sufficient.</p> <p>From the start date, you will have 1 year to complete your thesis. If you need longer and there are no extraordinary or extenuating circumstances, the project is considered a fail and you will need to find a new subject and supervisor.</p> <p>The thesis project is an individual project in which the student is challenged to apply the knowledge obtained by studies in the past years over a period of 9 months. This project is often performed together with or even at a maritime company.</p> <p>Start about 3 months in advance with the following:</p> <p>Contact the professor, or other academic staff members, of your specialization to see if there is a project available that you might like. TU Delft is frequently contacted by problem owners for such projects, but this is not guaranteed. In a discussion you can both find out what subject interests you and if there is a project available that relates to that subject. In most cases there is a rough problem description available of about ½ an A4. If not, most academic staff members can tell you where to search or who to contact to help find a suitable project.</p> <p>Another option is to contact a problem owner (company/PhD student/Staff Member) yourself (e.g. through your own network). If they agree to take you as a graduation student on a subject you find interesting, you should write a short description of about ½ an A4 to discuss with the specialization prof. This is usually sufficient to check if a problem is a viable subject to graduate on.</p> <p>Based on the initial description and discussions with all involved parties, you should extend the description to a more detailed Plan of Approach that ensures that both you and your supervisory team agree on what you will do in the next 9 months. If all parties agree with your Plan of Approach the initial phase is concluded and it is time to set the date for a kick-off meeting. Usually at this point, you should also be assigned a daily supervisor (the problem owner) and an academic supervisor (TUDelft).</p> <p>Detailed description of the process and graduation guidelines can be found on Brightspace.</p> |    |
| <b>Study Goals</b>                     | <p>The student:</p> <ul style="list-style-type: none"> <li>- Can independently collect, process and integrate theory from different fields or sources</li> <li>- Can explain and defend his ideas in a scientific article</li> <li>- Generate new knowledge or develop new designs</li> <li>- Can present his ideas clearly both in written form and in verbal form</li> <li>- Can defend his ideas in a debate with experts</li> <li>- Can be relied upon to manage and plan his own activities</li> <li>- Can set-up and perform experiments to support his ideas</li> <li>- Can critically reflect on his work</li> <li>- Can handle feedback in a positive way.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Education Method</b>                | Self study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Prerequisites</b>                   | At least 69 EC of the master program obtained, no obligatory courses open                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Assessment</b>                      | T1 Literature Research (written report)<br>T2 Presentation (pass/fail)<br>T3 Colloquium Attendance (pass/fail)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |

| MT54035                                | MT MSc Thesis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 35 |
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| <b>Responsible Instructor</b>          | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Education Period</b>                | None (Self Study)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Course Contents</b>                 | <p>This is part 2/2 of your master graduation project; it consists of the research solution, (numerical) experiments, gathering and critically reflecting on results and writing the thesis.</p> <p>The text below is the same as for MT54010.</p> <p>Before you can start your master thesis project you will need to hand in a signed MSc Thesis form (form 4) with the Director of Studies (P. de Vos). This form is available from the Master Marine Technology page on Brightspace. The form registers amongst others the working title of your thesis, your supervisor, a short description of your project, the start date and provisional graduation date. The form is signed by the Director of Studies only when your study progress as registered in OSIRIS is sufficient.</p> <p>From the start date, you will have 1 year to complete your thesis. If you need longer and there are no extraordinary or extenuating circumstances, the project is considered a fail and you will need to find a new subject and supervisor.</p> <p>The thesis project is an individual project in which the student is challenged to apply the knowledge obtained by studies in the past years over a period of 9 months. This project is often performed together with or even at a maritime company.</p> <p>Start about 3 months in advance with the following:</p> <p>Contact the professor, or other academic staff members, of your specialization to see if there is a project available that you might like. TU Delft is frequently contacted by problem owners for such projects, but this is not guaranteed. In a discussion you can both find out what subject interests you and if there is a project available that relates to that subject. In most cases there is a rough problem description available of about ½ an A4. If not, most academic staff members can tell you where to search or who to contact to help find a suitable project.</p> <p>Another option is to contact a problem owner (company/PhD student/Staff Member) yourself (e.g. through your own network). If they agree to take you as a graduation student on a subject you find interesting, you should write a short description of about ½ an A4 to discuss with the specialization prof. This is usually sufficient to check if a problem is a viable subject to graduate on.</p> <p>If a student undertakes the graduation project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-3mE graduation agreement should be used, which is available at the following link:<br/> <a href="https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/">https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/</a>.</p> <p>Based on the initial description and discussions with all involved parties, you should extend the description to a more detailed Plan of Approach that ensures that both you and your supervisory team agree on what you will do in the next 9 months. If all parties agree with your Plan of Approach the initial phase is concluded and it is time to set the date for a kick-off meeting. Usually at this point, you should also be assigned a daily supervisor (the problem owner) and an academic supervisor (TUDelft).</p> <p>Detailed description of the process and graduation guidelines can be found on Brightspace.</p> |    |
| <b>Study Goals</b>                     | <p>The student:</p> <ul style="list-style-type: none"> <li>- Can independently collect, process and integrate theory from different fields or sources</li> <li>- Can explain and defend his ideas in a scientific article</li> <li>- Generate new knowledge or develop new designs</li> <li>- Can present his ideas clearly both in written form and in verbal form</li> <li>- Can defend his ideas in a debate with experts</li> <li>- Can be relied upon to manage and plan his own activities</li> <li>- Can set-up and perform experiments to support his ideas</li> <li>- Can critically reflect on his work</li> <li>- Can handle feedback in a positive way.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Education Method</b>                | Individually Performed Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Prerequisites</b>                   | At least 69 EC of the master program obtained, no obligatory courses open                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Assessment</b>                      | <p>A written report (to be handed in 3 weeks before graduation) +</p> <p>An oral presentation (in English) +</p> <p>An oral exam a.k.a. the defense of your thesis in front of a graduation committee.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |

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|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Master Marine Technology</b>                       |

|                                                                                   |
|-----------------------------------------------------------------------------------|
| <b>MT Project / Electives (choose 1 out of 2, or 15 ECTS of elective courses)</b> |
|-----------------------------------------------------------------------------------|



| MT54000                                | Individual Research Assignment for MT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Dr.ir. P. de Vos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Education Period</b>                | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Course Contents</b>                 | In this course students will learn how to conduct a research assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Study Goals</b>                     | <p>After this course, the student is able to:</p> <ul style="list-style-type: none"> <li>- plan</li> <li>- carry out</li> <li>- report</li> </ul> <p>a research assignment by himself or herself.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Education Method</b>                | Selfstudy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Assessment</b>                      | <p>Written report on the research assignment that is carried out by the student.</p> <p>The report should comply with the rules for scientific writing (structure, references, language). Depending on the assignment, the report should clearly communicate the approach followed during the assignment, the results and findings obtained in a scientific manner.</p> <p>The grade will be based on the output of</p> <ul style="list-style-type: none"> <li>- (scientific) reporting skills</li> <li>- general research / engineering skills</li> <li>- technical expertise</li> <li>- social and communication skills</li> </ul> |    |
| <b>Remarks</b>                         | The individual research assignment cannot be used to do a regular internship. A scientific staff member of Maritime Technology is always first supervisor and he/she is responsible for the academic quality of the research assignment.                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Department</b>                      | 3mE Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |

| TUD4040                                | Joint Interdisciplinary Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Prof.dr.ir. J. Hellendoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Project Coordinator</b>             | Ir. B.J.E. de Bruin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Summary</b>                         | <p>JIP consists of three sets of activities:</p> <ol style="list-style-type: none"> <li>1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log.</li> <li>2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved.</li> <li>3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |
| <b>Course Contents</b>                 | <p>The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. Cognitive abilities attributable to interdisciplinary learning<br/>Demonstrate the ability to engage in perspective-taking;<br/>Develop structural knowledge pertaining to the problem;<br/>Integrate knowledge and modes of thinking drawn from two or more disciplines;<br/>Produce an interdisciplinary understanding of complex problem or intellectual question.</li> <li>2. Scientific and intellectual development<br/>Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation;</li> <li>3. Research and design capabilities<br/>Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute;<br/>Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design;</li> <li>4. Collaboration and communication in an interdisciplinary team<br/>Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures);<br/>Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages);<br/>Present work performed in a structured way through an oral presentation to their peers and customer.</li> <li>5. Self-adjustment and reflection capabilities<br/>Plan and control the project efficiently considering resources and methodology;<br/>Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.</li> </ol> <p>Cognitive abilities attributable to interdisciplinary learning:<br/>The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems<br/>Scientific and intellectual development<br/>The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation<br/>Research and design capabilities<br/>The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines<br/>Collaboration and communication in an interdisciplinary team<br/>Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders<br/>Self-adjustment and reflection capabilities<br/>To carry out regular reflections on professional and personal development and being able to improve upon those reflections<br/>Understand contemporary and societal issues in their work.</p> |    |
| <b>Education Method</b>                | <p>Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Assessment</b>                      | <p>The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level):<br/>Interdisciplinary work<br/>Scientific reasoning and ethical mindset<br/>Innovation process<br/>Presentation and communication</p> <p>Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review.<br/>The final group mark for JIP is differentiated per person and is based on:<br/>The team presentations - 30%<br/>The Problem definition, progress report and final report 30%<br/>The individual contribution to the team 20%<br/>The final individual reflection report 5%<br/>The Blog 5%</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Remarks</b>                         | <p>Project Outcomes (product, model, system) are presented at:</p> <ol style="list-style-type: none"> <li>1. Problem Statement Review</li> <li>2. Midterm meeting</li> <li>3. Final Review</li> <li>4. Symposium presentation</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |

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Ir. J.S. Grevink

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Dr. J.B. van Grunsven

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X. Jiang

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